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1. Title of the Invention

Pairwise-Interaction Sensor Fusion System with Protected Adaptive Statistics for Sub-Threshold Compound Hazard Detection and Graduated Automated Mitigation

2. Field / Area of Invention

This invention belongs to the fields of embedded systems engineering, real-time environmental safety instrumentation and computational sensor fusion. It describes an apparatus and a method for identifying developing hazards in enclosed laboratory spaces by fusing readings from physically dissimilar sensors into a single scalar severity measure, and for driving ventilation, alarm and indication hardware at a level graduated by the adopted band.

Teaching and research laboratories concentrate several distinct hazard sources into one enclosed volume. Solvent vapours, reactive gases, open flames, heat-generating instruments and electrical apparatus operate within metres of one another, yet the safety instrumentation installed in such rooms is normally a collection of independent devices, each carrying a threshold fixed at manufacture and none exchanging information with the others.

Two consequences follow. The first is that a hazard distributed across several parameters escapes notice entirely. An exothermic reaction beginning in a fume hood raises solvent vapour concentration and bench temperature together, and may bring both to sixty or seventy per cent of their respective alarm points without tripping either one. Each reading alone is unremarkable. Taken together, the pair is a recognizable precursor to a runaway reaction, and no instrument in the room performs that joint evaluation.

The second consequence concerns the fixed threshold itself. Solar gain, occupancy, fume-hood duty cycle and the heat rejected by nearby instruments move the resting condition of every channel across a working day. A threshold chosen for one operating point becomes either a nuisance-alarm generator or an insensitive detector as the room drifts away from it, and recalibration is manual and therefore performed rarely.

The present invention addresses both deficiencies within a single microcontroller-based unit. Per-channel protected baselines and dispersion estimates convert readings expressed in degrees Celsius, parts per million, percentage relative humidity and lux onto one common statistical scale. A pairwise interaction index then combines those standardized values, weighting parameter pairs by the separation measured during commissioning. The resulting scalar drives a three-band actuator response, while per-channel deviation bands and hard-coded absolute comparators operate as parallel decision paths.

2.1 Technical Problem Addressed by the Invention

- Independent single-parameter detectors cannot infer a hazard from the joint behaviour of several sub-threshold readings, so compound events remain undetected until one parameter reaches its individual alarm point.
- Fixed detection thresholds do not track slow environmental drift, producing nuisance activations as the room warms and reduced sensitivity as the resting condition rises.
- Readings from dissimilar transducers are expressed in incompatible units and cannot be combined arithmetically without an intermediate normalization stage. Normalization of heterogeneous sensor inputs onto a common scale is itself known, as Table 1 records; the difficulty addressed here concerns how the resulting standardized values are protected and combined, rather than normalization itself.
- Binary alarm outputs force an all-or-nothing response, with no mechanism for engaging partial ventilation while a condition is still developing.
- Purely statistical anomaly detection removes the absolute safety floor, allowing a slowly rising baseline to mask a genuinely dangerous absolute reading.
- Fusion schemes based on trained classifiers may require labelled training datasets and additional computational resources, motivating an explicit deterministic formulation suitable for low-cost embedded implementation.

2.2 Definitions

Composite Hazard Index (CHI). A scalar composite hazard measure computed each control cycle as the weighted sum of individual normalized deviations plus the sum of weighted pairwise products of those deviations. In the preferred form of Equation 4, in which the individual weights and the pairwise coefficients are each non-negative, the measure is non-negative, continuous-valued and not constrained to a fixed 0 to 1 range, and is monotonically non-decreasing in every normalized deviation; non-negativity is a preferred property of that form and is claimed dependently.

Normalized deviation. The positive part of the signed difference between a current reading and its adaptive baseline, divided by the dispersion estimate for that channel, giving a dimensionless quantity expressed in units of sigma. The photometric channel yields two such deviations, one for falling illuminance and one for rising illuminance.

Protected statistics. A dispersion register whose updating is inhibited while an excursion is in progress, together with a baseline register whose updating is separately inhibited while a mitigation actuator is operating. The two inhibitions have different trigger conditions and different purposes: the first prevents a developing hazard from widening the tolerance band against which it is judged, the second prevents the system's own response from being learned as the resting condition.

Adaptive baseline. A per-channel running adaptive reference computed from recent readings, representing the prevailing resting condition of that channel. In the disclosed embodiment it is an Exponential Moving Average updated once per ten-second epoch; the exponentially weighted form is preferred and is claimed dependently, the independent claims requiring a running average only.

Pairwise interaction coefficient. A weight assigned to a specific parameter pair and fixed after commissioning, non-negative in the preferred embodiment and claimed as such dependently, derived from the measured separation between the joint elevation of that pair during staged hazard events and during benign disturbances. These coefficients are not statistical correlation coefficients and are not computed from the running covariance of the channels.

Absolute backstop. A hard-coded comparison against a fixed physical limit that acts independently of the index and independently of the adaptive baselines.

Three-path arbitration. The logical arrangement in which a response is asserted by whichever of three parallel paths calls for the higher band: the composite index band, the per-channel normalized-deviation band, or the absolute backstop comparator. No path can suppress another.

3. Prior Patents and Publications from Literature

3A. Prior Patent Literature

S. No.	Patent No. (Year)	Title	Comments
1	US5691703A (1997)	Multi-Signature Fire Detector	The closest identified patent. It combines the outputs of plural fire-signature detectors by multiplication or by addition, applies weighting coefficients to the combined quantity, retains individual per-detector threshold processing as a fallback path, derives reference values from signal history rather than from a purely fixed constant, and contemplates particulate, gas and temperature detector types. Product formation, weighting, a fallback individual path and history-derived references are therefore all present in this single reference, and the present invention cannot be distinguished from it on any of those grounds taken alone. It does not disclose division of each deviation by a per-channel dispersion estimate, an all-pairs interaction set across four or more heterogeneous measurands combined with a weighted individual sum in one expression, inhibition of dispersion or baseline updating during an excursion or during mitigation, or actuation graduated by the magnitude of the combined quantity.
2	US7142105B2 (2006)	Fire Alarm Algorithm Using Smoke and Gas Sensors	Computes rates of increase of smoke, carbon monoxide and carbon dioxide and alarms when a derivative signal reaches a stated multiple of the measured noise standard deviation over a history window, so a running standard deviation is already used across three channels as the yardstick for an unusual change. The channels are all combustion products, the decision remains a comparison of each derivative against its own limit, and no product of two channel quantities is formed. Graduated actuation, an independent absolute path and dispersion-freeze logic are absent.
3	US5079422A (1991)	Fire Detection System Using Spatially Cooperative Multi-Sensor Input Technique	Maintains historical statistics of sensor behaviour, forms the difference between the current reading and a recent mean, estimates a standard deviation of that history, and evaluates the deviation in relation to that estimate. Standardization of a sensor reading against its own recent mean and dispersion is accordingly not novel in fire detection, and no contribution is claimed here for that operation alone. The reference forms no pairwise products of standardized deviations drawn from dissimilar measurands, does not inhibit statistic updating during an excursion, and does not drive graduated physical mitigation.
4	US20190108739A1 (2019)	Fire Detection System	Establishes long-term averages for each of several sensor types including smoke, temperature, gas and humidity, sets upper and lower control limits by offsetting those averages by a statistical parameter such as a standard deviation, and aggregates weighted sensor abnormality into an alarm decision, with the express aim of tolerating normal occupant activity. The combination of heterogeneous environmental sensing, control limits derived from long-term averages and standard deviation, weighting, and an aggregated severity output is therefore disclosed. What is not disclosed is the formation of pairwise products between the standardized deviations, the freezing of the statistics while an abnormal condition is in progress, the parallel absolute comparator, or graduated actuation of ventilation hardware.
5	US7202794B2 (2007)	Flame Detection System	Subtracts a mean coefficient from each processed sensor value and divides by a standard deviation coefficient in order to place heterogeneous sensor inputs on a scalable common footing before further combination. This is the same normalization step used here, applied for the same stated purpose of rendering dissimilar inputs comparable. The reference derives its scaling coefficients from a training set and passes the result to a trained neural network. In the present invention the baseline and dispersion state is maintained continuously from the installed environment, while the interaction coefficients are fixed after commissioning, and the combination is by an explicit and inspectable expression rather than a trained model. No interaction products, no absolute backstop path and no mitigation are disclosed.
6	US4665390A (1987)	Fire Sensor Statistical Discriminator	Discloses a cross-correlation detector operating in parallel with statistical discriminator circuitry, the outputs of the two routes being combined through decision logic, so parallel operation of a statistical path and a correlation path with a combined output is known and bears directly on the arbitration arrangement claimed here. The parallel paths operate on two radiation channels of one modality, the discriminators are not per-channel adaptive baselines maintained from the installed environment, and the second path is not a comparison against absolute physical limits. Neither graduated actuation nor protection of the statistics appears.
7	US5592147A (1997)	False Alarm Resistant Fire Detector with Improved Performance	Combines a carbon dioxide module with a further detector module by conjunctive confirmation and by disjunctive combination with an override route, so mixed arbitration between detection routes is disclosed. Both routes remain fixed-threshold fire detectors of one domain; no reading is standardized against its own dispersion and no severity magnitude is produced.

S. No.	Patent No. (Year)	Title	Comments
8	US5767776A (1998)	Fire Detector	Uses the relationship between a smoke channel and an infrared carbon dioxide channel, alarming on rate of change with the smoke channel qualifying the gas channel, so inter-channel qualification is known. The qualification is logical rather than multiplicative, both channels are combustion signatures, and thresholds are preselected.
9	US20050110632A1 (2005)	Multi-Sensor Fire Detectors with Audio Sensors and Systems Thereof	Adds acoustic and occupancy information to flame, smoke and temperature sensing and alters detector sensitivity accordingly, so context-dependent adjustment of a detection boundary is disclosed. The adjustment answers a discrete contextual input rather than a continuously maintained statistical description of each channel, and no composite magnitude or graduated mitigation appears.
10	CN102645908A (2012)	Gas-Leakage Emergency Protection Device	A programmable logic controller reads a carbon monoxide sensor and drives an alarm, a lamp and an exhaust fan when a fixed threshold is passed. The closest prior art in the sense-decide-ventilate chain, with a comparable actuator set. The decision rests on one parameter against one constant, with no second channel, no fusion and no baseline adaptation, and the fan is switched rather than modulated.
11	EP0440828A1 (1991)	Gas Monitoring System with Leak Detection and Flow Cutoff	Detects combustible gas and initiates a protective shut-off before a dangerous concentration is reached, with provision for continued supply under emergency conditions, so automatic protective actuation from a gas measurement and an override concept comparable to the backstop path are established. The measurement is single-parameter; cross-parameter inference, standardization and severity-graduated actuation are absent.
12	US5261268A (1993)	Gas Leak Detection System	Infers leakage from flow and pressure sensing in the supply passage by comparison against behaviour expected during ordinary consumption, and may operate a shut-off valve. The compared quantities describe the supply line rather than the ambient condition of the protected room, the expectation derives from consumption models rather than running statistics of the installation, and no severity magnitude is formed.
13	WO2024240674A1 / EP4713697A1	Method and Device for Anomaly Event Detection	The closest identified art to the statistic-protection feature. Detects anomaly events in an electrical system by forming the residual between a data point and an exponentially weighted moving average baseline, comparing that residual against an adaptive threshold formed as a mean variation multiplied by a boundary parameter, and counting exceedances in a leaky integrator. A flag designated Freeze is set on event detection and, while set, prevents residual values from entering the buffer from which the mean variation and hence the threshold are calculated, expressly so that oscillations during and after an event do not corrupt the threshold. Inhibition of a statistic used for threshold estimation during and after an event is therefore disclosed, and no novelty is asserted here in that act taken alone. The reference monitors a single variable of one physical kind, being a current, voltage, power or harmonic energy. It does not standardize a plurality of heterogeneous measurands by their respective dispersions, does not form pairwise products of such standardized deviations, does not arbitrate a composite magnitude against per-channel bands and absolute limits, and does not operate graduated ventilation hardware. Its inhibition protects a threshold buffer; the inhibition claimed here additionally extends to the baseline register during mitigation, closing the path by which the system's own response would otherwise be learned as the resting condition.
14	US4639598A	Fire Sensor Cross-Correlator Circuit and Method	Cross-correlates the signals of two fire detectors by sampling the paired signals, multiplying them and accumulating the products so as to confirm that both are responding to one source. Formation of a product of two detector signals is therefore old and is material to the pairwise product relied upon here. The correlated signals are two channels of one radiation-sensing modality, the product confirms common origin logically rather than contributing graded severity, and no operand is divided by a running dispersion estimate. No all-pairs set across dissimilar measurands, no protection of the reference statistics and no mitigation is disclosed.
15	US9218232B2 / US20140149806A1	Anomaly Detection Methods, Devices and Systems	Determines which of multiple sensors are correlated over a sliding history window, applies a normalization function stated to comprise a Z-transformation to the readings, and declares an anomaly where a deviation measure exceeds a threshold. Standardization of heterogeneous sensor readings onto a common sigma scale for lightweight online anomaly detection is therefore old, and is acknowledged as such here. The correlation is used to select which sensors are jointly considered and the deviation is an aggregate distance measure; no weighted product of two standardized deviations is formed, no statistic is protected against the event or the response, and no graduated actuation of mitigation hardware is disclosed.
16	US10503144B2 / US20190121317A1	Anomaly Detection with Correlation Coefficients	The most material reference to the pairwise element. Receives data from a plurality of sensors each monitoring a different process metric, defines two sensors as a sensor pair, generates a reference correlation for the pair, z-normalizes the difference of the remaining data from that reference,

S. No.	Patent No. (Year)	Title	Comments
			filters and identifies anomalies, and adjusts process parameters of the apparatus in response. Pair definition, z-normalization and physical parameter adjustment in response to a detected anomaly are therefore all disclosed together. The distinction relied upon is the quantity standardized: the reference z-normalizes the departure of a pair from its own reference correlation, whereas here each channel is independently standardized against its own baseline and dispersion and the standardized values are then multiplied, so that the product magnitude expresses joint elevation of two channels rather than departure from a correlation. The reference further discloses no protection of the statistics against the event or the mitigation, no arbitration against per-channel bands and absolute limits, and no graduated actuation of ventilation hardware.
17	US20220316730A1 / US11906182B2	Scoring a Building's Atmospheric Environment	The closest identified art to the composite-score architecture. Receives signals from heterogeneous sensors spanning temperature, humidity, carbon dioxide, particulate, total volatile organic compound and radon measurement, converts each dissimilar metric to a normalized unitless score by a bilinear mapping anchored at acceptable, enhanced and warning levels drawn from published building standards, combines the normalized scores into a single composite score by a weighted arithmetic or geometric average, compares the composite score and also the individual normalized scores against a common trigger threshold, and sends a signal to building equipment causing an increase in ventilation or in the supply of outdoor air. The chain running from heterogeneous environmental measurement through common normalization and weighted composite scoring to automatic ventilation remediation is therefore disclosed in one reference, and no novelty is asserted here in that chain as such. The normalization is a fixed mapping to externally published thresholds, not division of a baseline-referenced difference by a running per-channel dispersion maintained from the installed space; no product of two normalized values is formed, and because the composite is a weighted average it cannot exceed the range spanned by its own operands and so cannot reach an actionable level while every individual normalized score remains acceptable. The reference further discloses no inhibition of any statistic during an event or during the remedial response, no absolute comparator arbitrated as a third path against the composite and per-channel paths, and no actuator level selected from three or more levels according to an adopted band.
18	CN121264336A	Fungus and Wood-Ear Watering Control System	The closest identified art to the standardized interaction term. Discloses a cultivation-environment monitoring and control system in which four heterogeneous factors, being free-water duration, air temperature, air humidity and air flow rate, are each converted to a standard score by subtracting a reference mean and dividing by a reference standard deviation, those references being compiled from a historical database of previous production cycles and updated periodically. The standard scores are combined into a composite risk index containing a temperature-standard-score multiplied by humidity-standard-score interaction term whose coefficient is obtained by regression against historical outcome data, together with a sigmoid free-water term and an air-flow term that reduces the index, and the index is compared against a preset threshold to permit or prohibit operation of a solenoid valve. Multiplication of two standardized environmental variables to express a synergistic effect within a multi-factor index that gates an actuator is therefore disclosed, and no novelty is asserted here in that operation taken alone. The reference maintains no per-channel baseline or dispersion from live observation of the protected space, forms one interaction term rather than an all-pairs set, composes its terms multiplicatively rather than as a sum of weighted individual deviations and weighted pairwise products in one expression, inhibits no statistic during an event or during the response to it, arbitrates a single index against a single threshold rather than three parallel paths, and issues a binary permit or prohibit decision rather than selecting a level of physical mitigation graduated by an adopted band.
19	CN122217398A	Submarine Cable Joint Monitoring Device and Method	The closest identified art to the summation of pairwise products alongside individual terms. Discloses a joint monitoring device combining a structural strain unit, an electrical integrity unit and a microenvironment unit sensing temperature, relative humidity, air pressure and target gas concentration, in which every monitored parameter is reduced to an offset degree by dividing its departure from a normal baseline by the permitted fluctuation range, the offsets are fused within each unit, a synergistic amplification factor is applied where humidity and gas are abnormal together, and a health score is formed by adding to a weighted sum of the three unit offsets a collaborative penalty comprising the sum of the pairwise products of those offsets, the score being mapped to four health levels. Addition of pairwise products of normalized magnitudes to a weighted sum of the same magnitudes, expressly to represent the additional risk of simultaneous abnormality, is therefore disclosed, and no novelty is asserted here in that operation

S. No.	Patent No. (Year)	Title	Comments
			taken alone. The normalizing denominator is a fixed permitted range about a factory or standard baseline and not a dispersion estimate maintained from live observation of the protected space; no statistic is inhibited during an excursion or during a response; the products are formed between three aggregated unit scores obtained after intra-unit maximum operators rather than between per-channel standardized deviations of individual measurands in one expression; fault classification requires a trained neural network model; the rule engine feeds that classifier in cascade rather than three paths being arbitrated without mutual suppression, and one indicator is given veto power over its own unit; and the output is a bounded health score driving indicator lamps and maintenance recommendations, no mitigation actuator being commanded to a level.
20	WO2021257287A1 / US11579030B2 (Cirrus Logic, priority 18-06-2020, published 23-12-2021)	Baseline Estimation for Sensor System	Directly material to the baseline-inhibition feature taken alone. Discloses a baseline calculation engine that subtracts a running baseline from a sensor signal to yield a corrected signal, compares the resulting error against a threshold and reduces the tracking gain when it is exceeded, and expressly describes operation in which the baseline signal is frozen during a detected intentional human interaction so that a sustained input is not absorbed into the reference. Inhibition of an adaptive reference during the event it is intended to reveal is therefore established prior art, and no novelty is asserted here for statistic inhibition standing alone. The reference concerns a single force-sensing channel of a virtual button in a mobile device; there is no heterogeneous multi-domain sensing, no division by a per-channel running dispersion estimate, no pairwise product of standardized deviations, no separate inhibition of a dispersion statistic, no inhibition of any statistic in response to the system's own actuator state, no arbitration of parallel decision paths and no mitigation actuator of any kind. The freeze is triggered by the input itself; here the baseline freeze is triggered by the commanded state of the mitigation actuator, which is the feature no identified reference discloses.

Table 1: Prior Patent Literature

3B. Non-Patent Literature

S. No.	Source / Year	Title	Comments
1	Heskestad, G. and Newman, J. S., Fire Safety Journal, Vol. 18, No. 4, 1992, pp. 355-374	Fire Detection Using Cross-Correlations of Sensor Signals	The most material item of non-patent prior art. Products of the form of two and three sensor signals were evaluated for early fire detection, the products being averaged over a correlation interval, and every signal was referenced to its baseline average over a preceding interval. The pairings reported include a carbon dioxide with temperature product, which is the direct counterpart of the highest-weighted pair used here. Forming a product of baseline-referenced readings from dissimilar sensors is therefore established in the literature and is not claimed as novel in this disclosure. The reported products are time-averaged correlations of baseline-referenced differences that retain their physical units, so the operands are not commensurate and no common scale is established. There is no division by a per-channel dispersion estimate, no combination of an all-pairs product set with a weighted individual sum in a single expression, no inhibition of the reference statistics during an event, no absolute comparator operating in parallel, and no actuation of any kind. The distinction relied upon here is accordingly the protected standardization and the surrounding control architecture, not the act of multiplying sensor signals.
2	Roberts, S. W., Technometrics, Vol. 1, No. 3, 1959	Control Chart Tests Based on Geometric Moving Averages	Origin of the exponentially weighted moving average as a control statistic, establishing the recursion and the trade-off governed by the smoothing factor. Not claimed as novel. The treatment is univariate and the moving average is the decision reference for one characteristic, not a stage whose output becomes an operand for combination with others.
3	Montgomery, D. C., Wiley	Introduction to Statistical Quality Control	Standard treatment of exponentially weighted moving average charts, sigma-multiple control limits and smoothing-constant selection, supplying the justification for the two-sigma and three-and-a-half-sigma bands. Univariate throughout; multivariate charts test one aggregate statistic and neither weight individual variable pairs nor drive graduated actuation.
4	Gottuk, D. T. et al., Fire Safety Journal, 2002	Advanced Fire Detection Using Multi-Signature Alarm Algorithms	Experimental evaluation of alarm algorithms combining several fire signatures, quantifying the reduction in nuisance alarms relative to single-signature detection, and the strongest available evidence that combining signatures improves discrimination. All signatures originate from combustion, so it gives no guidance on combining channels whose units and dynamics differ by orders of magnitude.

S. No.	Source / Year	Title	Comments
5	Chen, Wong et al., Fire Safety Journal, 2007	Fire Detection Using Smoke and Gas Sensors	A staged algorithm in which an exceeded smoke rate gates examination of carbon monoxide and carbon dioxide rates. The gating is a form of inter-channel dependency expressed as a sequential logical test rather than an arithmetic product, so two channels each at seventy per cent of their gating rates produce no detection. Thresholds are fixed and no actuator response is graded.
6	Fonollosa, Solorzano and Marco, Sensors (MDPI), 2018	Chemical Sensor Systems and Associated Algorithms for Fire Detection: A Review	Surveys chemical sensing for fire detection and the pattern-recognition methods applied to sensor arrays, documenting the metal-oxide drift that motivates the adaptive baseline used here. The reviewed algorithms are predominantly trained classifiers needing labelled datasets and budgets beyond a low-cost microcontroller.
7	Hall, D. L. and Llinas, J., Proceedings of the IEEE, Vol. 85, No. 1, 1997	An Introduction to Multisensor Data Fusion	Foundational taxonomy of fusion architectures and of commensurability between heterogeneous sensor outputs, establishing that dissimilar sources must be brought to a common representation before combination. The principle is not reduced there to a realisation in which protected per-channel statistics supply that representation and pairwise products of the results form the decision quantity.
8	Espressif Systems	ESP32 Series Datasheet	Primary source for the processor, memory, converter and wireless capability of the selected controller, and the basis for the timing and memory budgets of Section 8. A component specification only.
9	Occupational Safety and Health Administration, 29 CFR 1910.1000, Table Z-1	Permissible Exposure Limits for Air Contaminants	Primary source for the 400 ppm eight-hour time-weighted average permissible exposure limit for isopropyl alcohol relied upon in Section 7.2, and the basis on which the absolute gas backstop of this disclosure is characterised as a conservative configured limit rather than a threshold of acute harm. A regulatory exposure limit only. It prescribes no detection method, no combination of measurements from dissimilar sensors and no actuation, and gives no guidance on a limit for a broadband metal-oxide element reporting an equivalent concentration against a calibration analyte.

Table 2: Non-Patent Literature

Note on citation verification. Every entry in Tables 1 and 2 carries a full identifier, being a patent publication number or a journal, volume and year, and each has been checked against the issuing patent office or publisher record. Titles, authors and dates are stated as recorded by that authority. Where a reference was located but its content could not be confirmed against the primary document, it has been omitted rather than characterised from a secondary source.

4. Summary and Background of the Invention

4.1 Practical Background and Limitations of Existing Solutions

A laboratory bench is a dense arrangement of independent hazards: reagent bottles beside a hot plate, a burner within reach of a solvent wash bottle, an instrument rack rejecting heat into the same enclosed air. Protection is normally delegated to devices bought separately and installed separately, each responsible for one variable and each unaware of the others.

That arrangement performs acceptably against a single dominant failure. A large leak reaches the gas detector's alarm point and an established fire reaches the heat detector. It performs poorly against the more common early condition in which several variables move a moderate distance together. Solvent evaporating from a warm spill raises vapour concentration and humidity while the bench warms, and a reaction beginning to run away raises temperature and releases gas. In each case every reading sits inside its permitted range and the installed instrumentation is silent by design.

Fixed thresholds compound the problem. The resting condition of a working laboratory moves by several degrees and by tens of parts per million across a day, so a threshold sensitive in the morning generates nuisance activations by mid-afternoon and one that survives the afternoon is insensitive in the morning. Operators respond by raising thresholds or disabling devices, which removes protection altogether.

4.2 Distinctions Over the Closest Reported Systems

The building blocks of this invention are individually old, and the disclosure does not assert otherwise. Products of baseline-referenced sensor signals were reported by Heskestad and Newman in 1992. Standardization of a reading against its own recent mean and dispersion appears in US5079422A and, expressly to make heterogeneous inputs comparable, in US7202794B2. Control limits at a standard-deviation offset from long-term averages across smoke, temperature, gas and humidity appear in US20190108739A1. Combination of detector outputs by multiplication with weighting and a

retained individual fallback appears in US5691703A, parallel operation of a statistical and a correlation path in US4665390A, and driving a fan from a gas reading in CN102645908A. Normalization of heterogeneous environmental measurements onto a common scale, weighted combination into a composite score, comparison of both the composite and the individual values against thresholds, and automatic operation of ventilation equipment appear together in US20220316730A1. A standardized temperature-by-humidity interaction term gating an actuator appears in CN121264336A; a sum of pairwise products added to a weighted individual sum appears in CN122217398A; and inhibition of an adaptive baseline during the event it is intended to reveal appears in WO2021257287A1.

The distinctions therefore cannot rest on any of those operations. Table 3 maps the resulting feature set against three of the closest identified references, and five differences are relied upon.

First, the operands are made commensurate before they are multiplied. In Heskestad and Newman the products are formed from baseline-referenced differences that retain their physical units, so a product of a concentration difference and a temperature difference carries the composite unit and its magnitude depends on the choice of units. Division by a per-channel dispersion estimate removes the unit, so that a product of two normalized deviations is a pure number whose magnitude means the same thing whichever pair produced it. This is what permits ten pairs drawn from five dissimilar measures to be summed against one another and against a weighted individual sum in a single expression.

Second, the reference statistics are protected. Dispersion updating is inhibited from the moment any channel exceeds one sigma until all channels return, and baseline updating is inhibited while mitigation is running. Inhibition of an adaptive reference during the event it is intended to reveal is itself known, WO2021257287A1 freezing a baseline during a detected interaction, and the Eaton reference blocking event-related values from an adaptive threshold buffer; no novelty is asserted for that act standing alone, the feature relied upon being inhibition of the baseline in response to the system's own commanded actuator state together with separate inhibition of a dispersion statistic. Neither US5079422A, US20190108739A1 nor US7202794B2 inhibits its statistics in this way. Without the inhibition a developing hazard widens the denominator against which it is judged, and an activated fan is learned as the new resting condition.

Third, three decision paths operate in parallel rather than two. US4665390A places a statistical path beside a correlation path, and US5592147A combines two fixed-threshold fire modules. Here the composite magnitude, the per-channel normalized-deviation bands and the absolute physical comparators are arbitrated together, so that adaptive sensitivity, per-channel protection and an absolute floor coexist and none can be suppressed by another.

Fourth, the output is a magnitude that sets the degree of physical mitigation. US5691703A produces an alarm indication, US20190108739A1 an alarm decision, and CN102645908A a switched fan. None of them modulates ventilation duty and exhaust aperture by mapping a continuous-valued severity measure to graduated actuator levels. US20220316730A1 sends a single remedial signal to building equipment, CN121264336A issues a binary permit or prohibit decision to a valve, and CN122217398A returns a four-level health score driving indicator lamps and maintenance recommendations without commanding any mitigation actuator.

Fifth, none of the three composite-index references produces the detection condition relied upon here, and Table 4 sets out the comparison feature by feature. US20220316730A1 combines its normalized scores by a weighted average, so the composite value cannot depart from the range spanned by its own operands and cannot reach an actionable level while every individual score remains acceptable, which is precisely the condition this invention exists to resolve. CN121264336A standardizes its four factors and forms one temperature-by-humidity product, but inside a multiplicative risk expression rather than an all-pairs set summed with a weighted individual sum. CN122217398A adds a sum of pairwise products to a weighted individual sum, but its operands are aggregated unit scores taken after intra-unit maximum operators rather than per-channel standardized deviations of individual measurands. In all three the reference statistics are fixed or historical rather than maintained from live observation of the protected space, no statistic is inhibited during an excursion or during the response

to it, no three paths are arbitrated without mutual suppression, and no mitigation actuator is commanded to a level graduated by the composite magnitude.

4.3 Inventive Combination and Cooperative Arrangement

The contribution is a cooperative arrangement, not any single operation within it. Exponentially weighted averages date from 1959, standardization against a running dispersion is present in the fire-detection art, products of baseline-referenced sensor signals were published in 1992, and driving a fan from a gas reading is older still. Each is acknowledged in Section 3.

What the identified art does not contain is the arrangement in which these elements condition one another. The exponentially weighted baseline and its dispersion estimate are positioned here as a unit-stripping stage whose output is an operand rather than a decision. That repositioning is what makes ten products across five dissimilar measurands mutually comparable and summable with a weighted individual term in one expression. Heskestad and Newman reported two hand-selected pairings of baseline-referenced signals rather than the standardized multi-pair structure used here.

The products then supply detection capability that the individual terms cannot. For the primary compound condition the ten interaction terms contribute 1.215 against 1.122 from the five individual terms, as computed in Table 9. For the weakest compound condition the individual sum alone is 0.82, below the warning boundary of 1.0, while the complete index reaches 1.45.

The protection of the statistics closes the loop. Because the baselines move, what counts as sub-threshold is itself time-varying, so the products are evaluated against a reference frame that tracks the room. That frame is protected in two distinct ways with two distinct triggers: the dispersion registers stop updating once any channel passes one sigma, so the developing event cannot widen the band against which it is judged; and the baseline registers stop updating whenever an actuator is commanded, so the response cannot be learned as the resting condition. Neither inhibition applies to both registers, and neither trigger applies to both conditions. A fixed-threshold product has a static frame, and an adaptive frame left unprotected in either respect is corrupted by the very conditions it must resolve. The ablation reported in Section 8.6 shows that removing either element leaves the observed detection characteristic unreproduced.

4.4 Mapping of Key Features Against Closest Prior Art

Technical feature	Present invention	US5691703A	US20190108739A1	Heskestad 1992
Channels combined	5 heterogeneous measurands	Plural fire signatures	Smoke, temperature, gas, humidity	2 to 3 fire signatures
Standardization by running dispersion	Yes, per channel	No	Control limits at a sigma offset	No, units retained
Operands commensurate before multiplication	Yes	No	Not multiplied	No
Products formed	All 10 pairs, instantaneous	Product of detector outputs	None	2 hand-selected, time-averaged
Dispersion held during excursion; baseline held during mitigation	Yes, under separate trigger conditions	No	No	No
Parallel absolute comparator	Yes, third path	Individual fallback thresholds	No	No
Actuation	Graduated, duty and aperture	Alarm indication	Alarm decision	None

Table 3: Feature Mapping Against the Closest Identified Prior Art

Technical feature	Present invention	US20220316730A1	CN121264336A	CN122217398A
Reference statistics	Running per-channel baseline and dispersion from the installed room	Fixed bilinear mapping to published building standards	Batch mean and SD from a historical production database	Fixed permitted range about a factory baseline
Interaction terms	A plurality of pairwise products; the five-channel embodiment forms all ten	None; scores averaged	One temperature-by-humidity product	Pairwise products of 3 aggregated unit scores

Technical feature	Present invention	US20220316730A1	CN121264336A	CN122217398A
Form of combination	Weighted individual terms and weighted pairwise products in one sum	Weighted arithmetic or geometric average	Multiplicative risk expression	Weighted unit sum plus penalty, after intra-unit max
Action while all channels sub-threshold	Yes; index 2.95 at max 1.97 sigma	No; average bounded by its operands	One index, one threshold	Advisory only; no actuation
Statistic inhibition	Dispersion held during excursion; baseline held during mitigation	No	No	No
Decision structure	Three parallel paths, none able to suppress another	Composite and individual scores, common trigger	Single index, single threshold	Rule engine cascading into a trained classifier; one indicator holds veto
Output	Ventilation duty and exhaust aperture, three levels	Single remedial ventilation signal	Binary permit or prohibit to a valve	Four-level health score, lamps and advice

Table 4: Feature Mapping Against the Composite-Index and Interaction-Term References

Table 3 maps the closest fire-domain references. Table 4 maps the three composite-index and interaction-term references, being US20220316730A1, CN121264336A and CN122217398A, which are entered at rows 17 to 19 of Table 1 and are the references most material to the fusion expression of Equation 4. They are distinguished in the fifth paragraph above and in Section 9.2.

4.5 Technical Effects Achieved

The arrangement produces measurable effects, each traced to a specific mechanism and each quantified in Section 8.

Compound events that reach no individual threshold are detected. Across three staged compound conditions the index reached 1.45, 2.34 and 2.95 while the highest normalized deviation on any channel was 1.97 sigma, below the 2 sigma individual boundary, and a fixed-threshold configuration produced no output for any of the three. Detection performance over the sixty-event set improved from 51.67 % to 96.67 % accuracy and from 50.85 % to 97.50 % F1 score, with recall rising by 160.0 % in relative terms and false positives falling from four to one. These figures were obtained in one room with coefficients commissioned in that room and are not offered as validated performance across installations.

Nuisance activation under environmental drift was eliminated. A four-hour run with 6 °C of ambient drift held the fixed 35 °C threshold in a false warning state for 8.3 % of the run, while the adaptive boundary recorded no false warning at any magnitude tested up to 10 °C.

Automatic actuation removes the operator response dead time. Mean end-to-end response delay across four event types was 1.96 s, and automatic ventilation returned the space from 420 ppm to below 200 ppm in 253.7 s against 372.6 s for manual response. The two ventilated cases share the same decay time constant, so this advantage is attributable to the elimination of dead time and not to any intrinsic effect of the fusion stage on gas removal. The computational cost stays inside an inexpensive controller: worst-case fusion time was 48 ms, which is 9.6 % of the 500 ms cycle and 2.4 % of the mean response delay, with heap occupancy at 68.1 %.

4.6 Industrial and Practical Applicability

Teaching and research chemistry laboratories are the primary application, where solvents, burners and heated apparatus create the multi-parameter precursor conditions the index is built to catch, and where budget constraints favour a single low-cost unit over an instrumented safety system. Fume hoods and glove boxes present a smaller enclosed volume in which parameters couple strongly, so interaction terms grow quickly and warning-band ventilation can be engaged well before any absolute limit is approached.

Chemical and pharmaceutical storage rooms benefit chiefly from baseline adaptation, since these spaces drift widely with season and door traffic. Battery testing and charging areas exhibit a joint rise of temperature and off-gassing during thermal runaway, which maps onto the pair carrying the largest coefficient. School laboratories gain a supervision aid where staffing is thin, and small workshops handling adhesives, solvents and hot work apply the same unit by re-running commissioning.

5. Objective(s) of the Invention

The primary objective is to provide a single embedded unit that detects hazards distributed across several environmental parameters at magnitudes no individual detector would report, that maintains its sensitivity as the installed environment drifts, and that mitigates the detected condition at a level graduated by the adopted band without surrendering the absolute protection offered by conventional fixed limits. The specific objectives follow.

- To acquire readings from five heterogeneous channels spanning chemical, thermal, hygrometric, optical and photometric domains at their transducer-limited rates and evaluate them on a common 500 ms cycle.
- To maintain for each channel an Exponential Moving Average baseline and a dispersion estimate over a three-hundred-second window, converting every reading to a dimensionless deviation on a common scale, to inhibit updating of the dispersion registers while an excursion is in progress, and separately to inhibit updating of the baseline registers while a mitigation action is in progress.
- To compute a Composite Hazard Index each control cycle as the weighted sum of the five standardized deviations plus ten pairwise interaction products, the coefficients being derived by the commissioning rule of Section 7.4 from the separation measured between staged hazard events and benign disturbances.
- To detect compound hazards in which every individual channel remains below its own warning boundary, and to demonstrate that condition quantitatively against a fixed-threshold reference on a common event set.
- To classify the index into safe, warning and critical bands and to drive ventilation, vent angle, audible alarm and visual indication in graduated steps, so that a developing condition receives partial mitigation before it becomes acute.
- To retain per-channel deviation bands and absolute backstop comparators as parallel decision paths, arbitrated so the highest band called for by any path prevails and none can suppress another.
- To hold the fusion computation and the complete control cycle inside the resources of a low-cost microcontroller, with the fusion stage occupying a small and measured fraction of the cycle budget.
- To transmit raw readings, baselines, normalized deviations, index values and actuator states to a persistent store and a live interface, so that the reasoning behind every actuation can be reconstructed afterwards.
- To establish the pairwise coefficients by an explicit and reproducible commissioning rule executed in the target room, so that detection behaviour reflects the installation rather than a factory default.

6. Working Principle of the Invention (in brief)

The unit executes one closed cycle every 500 milliseconds. The cycle proceeds through sensing, baseline maintenance, normalization, fusion, arbitration, actuation and transmission, and every stage completes before the next begins. Table 5 separates the combination for which protection is sought from the implementation choices that may be replaced.

Sensing. Five channels are read at their individual maximum update rates. An MQ135 metal-oxide element reports gas concentration through the analogue converter at 2 Hz, an infrared flame sensor reports combustion-band activity as a duty fraction at 2 Hz, and a light-dependent resistor reports illuminance at 2 Hz. A DHT11 reports temperature and relative humidity at 0.5 Hz, which is the fastest rate that device supports, and the most recent value is held between updates. The channels were chosen for dissimilarity rather than redundancy, because heterogeneous measurands provide complementary information about distinct manifestations of a compound hazard.

Baseline maintenance. Each channel carries its own baseline register, updated once per ten-second epoch from the mean of the samples acquired in that epoch, according to $B(t) = a \times S(t) + (1 - a) \times B(t-1)$ with $a = 0.05$. A dispersion register holds the standard deviation over a trailing window of three hundred seconds, which is six hundred samples on the 2 Hz channels and one hundred and fifty on the

0.5 Hz channels. Both registers are protected: dispersion updating is inhibited for the duration of an active excursion, and baseline updating is additionally inhibited while mitigation is running.

Normalization. Each reading becomes n(t) = max(0, (S(t) - B(t)) / sigma(t)). The photometric channel is treated in both directions and yields two deviations from one transducer: an obscuration deviation using (B - S), which carries the channel weight because dense vapour or smoke reduces illuminance, and an emission deviation using (S - B), which is the operand used in the flame-light pair because combustion also emits in the visible band. Without this separation a rising illuminance would clamp to zero and the flame-light term could never form. The clamp discards deviations in the non-hazard direction. What leaves this stage is a set of pure numbers on one scale. Physical units are gone, and multiplication between channels is now meaningful.

Fusion. The index is formed as CHI(t) = SUM over i of w_i x n_i(t), plus SUM over i less than j of c_ij x n_i(t) x n_j(t). The five weights total 1.00 and express the individual severity of each channel. The ten coefficients express the strength of the physical mechanism coupling each pair, the largest being 0.30 for gas and temperature, which move together during exothermic reaction, and 0.25 for gas and flame, the combination that precedes ignition of an accumulated vapour. A product term contributes appreciably only when both of its operands are elevated, which is exactly the compound signature that individual thresholds cannot express.

Arbitration. Three paths run in parallel. The index is classified as safe below 1.0, warning from 1.0 to below 3.0, and critical at 3.0 and above. Independently, the designated per-channel standardized deviation of each sensing channel is classified at 2 sigma for warning and 3.5 sigma for critical; where a channel yields a second deviation, as the photometric channel does, that second deviation serves as an interaction operand only and is not banded. Independently again, selected raw readings are compared against hard absolute limits of 400 ppm isopropanol-equivalent, 45 °C and sustained flame. The remaining fixed values of Table 11, being 35 °C, 200 ppm, 75 per cent relative humidity and 300 lux, belong to the fixed-threshold reference configuration against which the invention is compared in Section 8 and are not actuator-triggering paths of the invention; within the invention a warning is asserted only by the index reaching 1.0 or by a per-channel deviation reaching two sigma. The illuminance channel carries no absolute limit. The arbiter adopts the highest band called for by any path.

Actuation. The warning band brings the ventilation motor to 50 % duty, opens the servo vent to 45 degrees, pulses the buzzer at 1 Hz and turns the indicator amber. The critical band commands full duty, 90 degrees, continuous alarm and red indication.

Transmission and feedback. Raw readings, baselines, dispersion values, normalized deviations, the index, its component breakdown and the actuator state are packetized every two seconds and pushed over Wi-Fi to a persistent store and a live interface, while a local display shows current status. The actuator state returns to the baseline estimator, which suspends baseline updating while ventilation is running so that the mitigation does not train the reference toward the hazard condition.

6.1 What Is Being Invented Compared With One Example Implementation

Protectable functional and structural combination	Replaceable choices in the described embodiment
Acquisition from at least four heterogeneous channels spanning distinct physical domains	MQ135, DHT11, infrared flame sensor and LDR as the specific transducers
Per-channel running adaptive baseline and dispersion estimate acting as a unit-stripping normalization stage	Exponentially weighted form, smoothing factor 0.05, ten-second update epoch, three-hundred-second dispersion window
Composite index formed as a weighted sum of normalized deviations plus weighted pairwise products of those deviations	The particular weight vector and the particular ten coefficient values
Inhibition of dispersion updating while an excursion is active and of baseline updating while mitigation runs	Excursion entry criterion set at one sigma above baseline
Classification of the index into three or more bands with actuation graded by band	Band boundaries at 1.0 and 3.0
Three parallel decision paths arbitrated by adopting the highest band called for by any of them	Band boundaries at 1.0 and 3.0, per-channel bands at 2 and 3.5 sigma, limits of 400 ppm, 45 °C and sustained flame
Bidirectional treatment of the photometric channel yielding separate obscuration and emission deviations	ESP32 controller, 2 Hz cycle, Wi-Fi transport, SQLite store, browser interface
Commissioning procedure that establishes coefficients and baselines in the target room	Duration and composition of the commissioning event set

Table 5: Claimed Combination Compared With Replaceable Implementation Choices

6.2 Statement Regarding Hardware Implementation

The invention comprises a hazard detection and mitigation apparatus configured to monitor the physical state of an enclosed space and to operate ventilation and alarm hardware automatically. It is constituted from commercially available components: one or more processors, memory, sensor interfaces, a metal-oxide gas element (101), a hygrothermal transducer (102, 103), an infrared flame sensor (104), a photometric element (105), a motor driver and ventilation motor (401), a servo actuator (402), an audible alarm (403), a visual indicator (404) and a wireless interface (501). What distinguishes it is the arrangement of those components and the manner in which they act upon one another: statistical state derived from live observation of the protected space is held in registers whose updating is inhibited by the very conditions the apparatus is detecting and by its own response, and that protected state determines, within a bounded control cycle, the duty applied to the ventilation motor and the angular position of the servo actuator. The output is not a score, an alert or a report; it is a set of actuator commands that alter the physical condition of the room.

6.3 Technical Contribution

- The exponentially weighted baseline is repurposed from a decision reference into a normalization operator, and it is that repositioning, rather than the recursion, that makes cross-domain multiplication possible.
- Pairwise products of standardized deviations create a decision quantity with no counterpart in single-channel comparison, contributing 1.215 of the 2.34 index value in the primary compound case.
- Coefficients are derived by an explicit commissioning rule from the measured separation between staged hazard events and benign disturbances, which keeps the index inspectable and removes any trained-classifier requirement, while inhibiting dispersion updating during an excursion closes the failure mode in which a developing hazard widens its own tolerance band.
- Arbitration across a composite adaptive statistic, per-channel adaptive bands and absolute physical comparators gives adaptive sensitivity without giving up either single-parameter sensitivity or an absolute safety floor.
- A continuous-valued index not constrained to a fixed 0 to 1 range supports graduated actuation, so partial ventilation can be applied to a developing condition rather than waiting for a binary trip, and suspending baseline update during mitigation stops the control action being absorbed into the reference.
- Separating the photometric channel into obscuration and emission deviations allows one transducer to serve both an obscuration hazard and optical confirmation of flame, which a single clamped deviation cannot do.
- The complete arrangement executes in 48 ms of a 500 ms cycle on a controller costing a few dollars, which places compound-hazard detection within reach of installations that cannot support an instrumented safety system.

6.4 Brief Description of the Drawings

Figure 1 shows the overall architecture as four layers with reference numerals, from the five sensing channels (101 to 105) through the normalization registers (201 to 205), the fusion and arbitration elements (301 to 304), and the actuation and communication outputs (401 to 504). Figure 2 shows the assembled prototype. Figure 3 presents the ten pairwise interaction coefficients as a matrix. Figure 4 shows the live interface. Figure 5 sets out the allocation of one 500 ms control cycle. Figures 6 through 11 present the quantitative validation results of Section 8.

7. Description of the Invention in Detail

Scope of Implementation. The implementation described in this section uses an ESP32 controller, an MQ135 metal-oxide gas element, a DHT11 hygrothermal transducer, an infrared flame sensor, a light-dependent resistor, a Wi-Fi transport, an SQLite store and a browser interface as the platform for reduction to practice and performance evaluation. These are representative implementation choices and not mandatory elements of the invention. The system extends to any functionally equivalent

alternatives, including non-dispersive infrared gas elements, faster hygrothermal transducers, thermopile flame detectors, any controller offering the required interfaces and computational headroom, and any wired or wireless transport of comparable throughput. Substitution of any of these does not alter the arrangement claimed in Section 10.

7.1 System Scope and Embodiment

The invention is embodied as a self-contained wall or bench mounted unit comprising a microcontroller, five sensing channels, four actuation outputs, a local display and a wireless link. The unit is installed in the room it protects, is commissioned in that room, and thereafter operates without external computation. Figure 1 gives the layered architecture with the reference numerals used throughout this section, Table 6 lists the elements against those numerals, and Figure 2 shows the assembled prototype.

Processing is organized in four layers. The sensing layer (100) acquires readings. The normalization layer (200) maintains per-channel statistics and converts raw readings to dimensionless deviations. The fusion and decision layer (300) computes the index, classifies it and arbitrates against the absolute comparators. The actuation layer (400) and communication layer (500) execute and record the outcome.

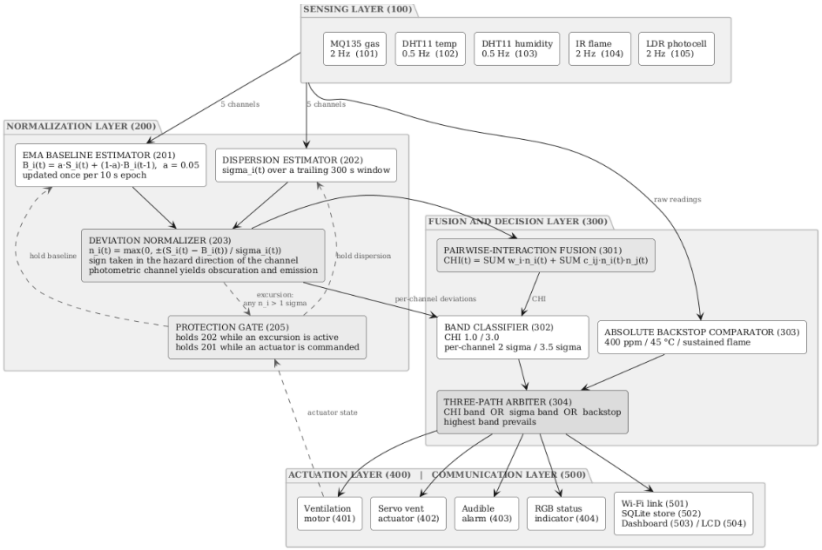


Figure 1: Layered System Architecture with Reference Numerals

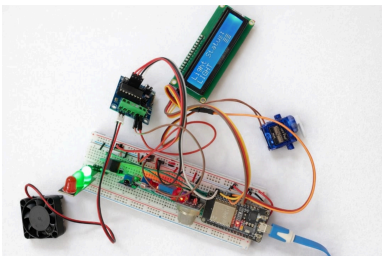


Figure 2: Assembled Prototype Unit

Numeral	Element	Specification and operating range	Function in the arrangement
101	MQ135 metal-oxide gas element	Broadband metal-oxide element, not species-selective; calibrated against isopropanol vapour and reported as ppm isopropanol-equivalent over 50 to 500 ppm, about 5 ppm resolution at midscale, 48 h preheat, sampled at 2 Hz	Chemical channel, index weight 0.35
102	DHT11 temperature channel	0 to 50 °C, manufacturer accuracy about ±2 °C, maximum update rate 0.5 Hz, value held between updates	Thermal channel, index weight 0.25

Numeral	Element	Specification and operating range	Function in the arrangement
103	DHT11 humidity channel	20 to 90 %RH, manufacturer accuracy about ±5 %RH, maximum update rate 0.5 Hz	Hygrometric channel, index weight 0.07
104	Infrared flame sensor	Combustion band, reported as duty fraction 0 to 1 over the trailing window, sampled at 2 Hz	Optical channel, index weight 0.25
105	Light-dependent resistor	0 to 1000 lux, about 5 lux resolution, sampled at 2 Hz	Photometric channel, weight 0.08 on the obscuration deviation; emission deviation feeds the flame pair only
201, 202	EMA baseline and dispersion estimators	Smoothing 0.05 per 10 s epoch; standard deviation over a trailing 300 s window, being 600 samples at 2 Hz and 150 at 0.5 Hz	Establish the resting condition of each channel and what change is unusual
203, 205	Deviation normalizer and statistic-protection gate	Clamped at zero, expressed in sigma; dispersion held from the one-sigma excursion entry until all channels return; baseline separately held while any actuator is commanded	Strips units, makes channels multiplicable, and stops the reference absorbing the event or the response
301	Pairwise-interaction fusion module	5 weighted terms plus 10 pairwise products, 48 ms worst case	Produces the composite severity magnitude
302, 303	Band classifier and absolute backstop comparator	Index bands 1.0 and 3.0, per-channel bands 2 and 3.5 sigma; limits 400 ppm, 45 °C, sustained flame	Map magnitude to a band, retain adaptive single-parameter bands, hold an absolute floor
304	Three-path arbiter	Adopts the highest band called for by the index, per-channel or backstop path	Combines composite, single-parameter and absolute protection without mutual suppression
401, 402	Ventilation motor with driver and servo vent actuator	0, 50 and 100 % duty via H-bridge; 0, 45 and 90 degrees	Remove accumulated vapour and heat with aperture graduated by the adopted band
403, 404	Audible alarm and RGB status indicator	Silent, 1 Hz pulse, continuous; green, amber, red	Signal the active band to occupants locally
501 to 504	Wi-Fi link, persistent store, browser interface, local display	2 s packet interval, 2 s interface refresh	Remote awareness and reconstruction of past events

Table 6: Component and Module Schedule

7.2 Sensing Layer (100)

The control cycle is 500 ms. Gas, flame and illuminance are sampled at 2 Hz. The DHT11 supplying temperature and relative humidity cannot be read faster than about 0.5 Hz, so those channels update every 2 s and the most recent value is held for intervening cycles, leaving two of the five operands piecewise constant between updates. This asymmetry is a property of the selected transducer, not of the method, and Section 7.9 identifies replacements that remove it. The set was assembled for physical dissimilarity: gas concentration responds to chemistry, temperature to energy release, humidity to phase change, the flame channel to radiant emission, and illuminance to obscuration or to the light of a flame.

The flame channel warrants a note. A bare infrared detector jumps between zero and one and cannot express a developing condition, which makes it unsuitable as a multiplicative operand; the unit therefore reports the duty fraction over the trailing window, a continuous quantity between zero and one. The gas channel warrants a stronger caveat. The MQ135 is a broadband metal-oxide element and is not species-selective, so a reading in parts per million is meaningful only against a stated calibration analyte. Every gas figure in this disclosure is expressed as ppm isopropanol-equivalent, established by a six-point calibration against a reference monitor after a 48 h preheat. The 200 ppm value is the fixed warning level of the reference configuration only, whereas 400 ppm is the absolute gas backstop, as Section 8.2 records. Neither is asserted as an acute toxicity limit: the Occupational Safety and Health Administration limit for isopropanol is 400 ppm as an eight-hour time-weighted average and the immediately dangerous concentration is several times higher, so the adopted value is a conservative backstop rather than a threshold of acute harm.

7.3 Normalization Layer (200)

Three registers are maintained per channel. The baseline register (201) holds B(t) and is updated once per ten-second epoch from the mean of the samples acquired within that epoch, using B(t) = a x S(t) + (1 - a) x B(t-1) with a = 0.05. The dispersion register (202) holds sigma(t), the standard deviation over a trailing window of three hundred seconds, which is six hundred samples on the 2 Hz channels and

one hundred and fifty on the 0.5 Hz channels. The normalizer (203) produces n(t) = max(0, (S(t) - B(t)) / sigma(t)).

Two design decisions in this layer carry weight. The first is the ten-second epoch. Updating the baseline at the full 2 Hz sample rate would let a fast excursion be absorbed into the reference within seconds, which is the classical failure of adaptive thresholding against a rising signal. Updating once per epoch gives an effective time constant of about 200 seconds per e-fold, so the baseline follows daily drift and ignores an event lasting a minute.

Equation 1: Exponentially Weighted Baseline Update (201)

$B_i(t) = a \times S_i(t) + (1 - a) \times B_i(t-1), \quad a = 0.05$

where S_i(t) is the mean of the samples acquired by channel i within the ten-second epoch ending at t, and B_i is the baseline register for that channel. The epoch is deliberately longer than the sampling interval, so that the reference tracks drift over hours while remaining insensitive to excursions lasting seconds.

Equation 2: Dispersion Estimate and Protection Condition (202, 205)

sigma_i(t) = standard deviation of channel i over a trailing window of 300 s

Update of sigma_i is inhibited for all i from the instant that any n_j(t) exceeds 1.0 until every n_j(t) returns below 1.0. Update of B_i is inhibited for all i while any mitigation actuator is commanded above its rest state. The window is fixed in time rather than in samples, so that channels of differing update rate share one reference horizon: 600 samples at 2 Hz and 150 at 0.5 Hz.

Equation 3: Standardized Deviation (203)

$n_i(t) = \max(0, (S_i(t) - B_i(t)) / sigma_i(t))$

The photometric channel yields two deviations from the one transducer:

$n_{obs}(t) = \max(0, (B_L(t) - S_L(t)) / sigma_L(t))$ obscuration, carries the channel weight

$n_{emis}(t) = \max(0, (S_L(t) - B_L(t)) / sigma_L(t))$ emission, operand of the flame pair only

At most one of n_obs and n_emis is non-zero at any instant.

The second is protecting the registers. If sigma grew as a hazard developed, the denominator would rise with the numerator and the deviation would flatten, so the hazard would widen the tolerance band against which it is judged. The dispersion register is therefore held at its pre-event value from the moment any channel exceeds one sigma until all channels return. The baseline register is separately held whenever a mitigation actuator is running, which prevents the ventilation response from being learned as the new resting condition. Inhibited baseline updating under a gas response is itself known in gas-detector art. What is put forward here is the joint protection of both registers within the arrangement of Section 10, not the inhibition alone.

The bidirectional treatment of the photometric channel set out under Equation 3 exists because a single clamped deviation cannot serve both purposes: were the channel clamped in the obscuration direction alone, a flame that raises illuminance would drive the deviation to zero and the flame-light term could never form. Table 7 collects the configuration values for this layer.

Parameter	Symbol	Value	Basis for the value
EMA smoothing factor	a	0.05	Gives about 200 s per e-fold at a 10 s epoch, tracking drift while ignoring minute-scale events
Baseline update epoch	T_e	10 s, mean of the samples in the epoch: 20 at 2 Hz, 5 at 0.5 Hz	Decouples reference adaptation from the 2 Hz decision rate
Dispersion window	T_w	300 s, being 600 samples at 2 Hz and 150 at 0.5 Hz	Fixed in time rather than in samples so channels of different rates share one reference horizon
Individual warning boundary	k_w	2 sigma	About the 97.7th percentile of a normal reference distribution for a one-sided positive deviation
Individual critical boundary	k_c	3.5 sigma	Extreme outlier, tolerant of the non-normal tails of low-cost sensors
Dispersion hold entry	1 sigma	Any channel above 1 sigma	Holds the dispersion estimate before the reference can be distorted

Parameter	Symbol	Value	Basis for the value
Baseline hold condition	Mitigation active	Any actuator commanded above its rest state	Prevents the control action being absorbed into the reference
Initialization period	T_i	10 min	Populates baselines and dispersion while absolute backstops carry protection
Channel update rates	f_i	2 Hz gas, flame, light; 0.5 Hz temperature, humidity	Set by the transducers; index evaluated at 2 Hz with held values
Dispersion floor	sigma_min	Per channel, from sensor resolution	Prevents division by a near-zero denominator in a very quiet room

Table 7: Adaptive Baseline and Dispersion Configuration

7.4 Pairwise-Interaction Fusion Layer (301)

The index is computed once per cycle as

CHI(t) = SUM over i of [w_i x n_i(t)] + SUM over i less than j of [c_ij x n_i(t) x n_j(t)]

The five weights sum to 1.00. The ten coefficients c_ij are non-negative constants fixed after commissioning. They are not statistical correlation coefficients and are not computed from the running covariance of the channels. The term used throughout this disclosure is pairwise interaction coefficient. Figure 3 presents them as a matrix.

Equation 4: Composite Hazard Index (301)

CHI(t) = SUM over i of [w_i x n_i(t)] + SUM over i less than j of [c_ij x n_i(t) x n_j(t)]

subject to w_i greater than or equal to zero for every i, SUM of w_i = 1.00, and c_ij greater than or equal to zero for every pair. The first summation expresses individual severity; the second expresses the joint elevation of physically related pairs and is the term that reaches an actionable value while every n_i remains below its own boundary. Non-negativity of both sets of weights, together with the clamping of each n_i at zero, makes the index monotonic: no channel can reduce the index by moving further into hazard, and no pair can offset another.

Equation 5: Commissioning Rule for the Interaction Coefficients

d_ij = median over staged hazard events of [n_i x n_j] - median over benign events of [n_i x n_j]

c_ij = round(c_max x d_ij / d_max , 2), c_ij set to zero where d_ij is at or below 0.05

where d_max is the largest separation observed and c_max is a scale factor determined by the ordered procedure below, such that a designated onset condition evaluates to the warning boundary of 1.0 within a stated tolerance after quantization.

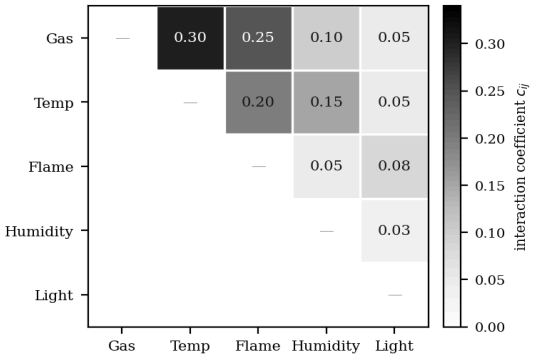


Figure 3: Pairwise Interaction Coefficient Matrix Over the Five Channels

Commissioning rule for the coefficients. The coefficients are produced by an explicit ordered procedure rather than assigned by judgement. For each pair (i, j) the commissioning set comprises at least five staged events of the hazard type that pair expresses, together with the twenty benign disturbance events, and the separation statistic and the zero-coefficient rule are those of Equation 5. An unquantized scale factor c_max is determined such that the designated onset condition evaluates to the warning boundary of 1.0; each coefficient is derived as c_max x d_ij / d_max, where d_max is the largest separation observed; the coefficients are quantized to two decimal places; and the onset condition is re-evaluated with the quantized set and accepted if it lies within 0.02 of the boundary, c_max being re-determined and the steps repeated where it does not. Quantization does not preserve exact equality, so the condition holds within that

tolerance. The designated onset condition is a separate commissioning case, distinct from the evaluation scenarios of Table 13: 138 ppm isopropanol-equivalent, 29.9 degrees Celsius, a flame duty fraction of 0.30, 52 %RH and 305 lux. It yields n_gas = 1.12, n_temp = 0.43, n_flame = 0.43, n_hum = 0.20 and an obscuration deviation of 0.23, giving an individual sum of 0.640 and an interaction sum of 0.360, hence CHI = 1.00. With c_max = 0.30 the accepted set of Table 8 places the onset at 1.00 to two decimal places, within tolerance. Scenario S7 returns 1.45. Acceptance further requires no activation over a four-hour benign run.

Pair (i, j)	d_ij	d_ij / d_max	c_ij	Candidate interaction associated with the configured hazard scenario
Gas and temperature	1.80	1.000	0.30	Staged exothermic reaction; incipient runaway
Gas and flame	1.50	0.833	0.25	Vapour with an ignition source; leak near a burner
Temperature and flame	1.20	0.667	0.20	Radiant emission with thermal escalation; early-stage fire
Temperature and humidity	0.90	0.500	0.15	Staged evaporation from a heated aqueous system. Relative humidity may instead fall with temperature where no moisture is released, so this pair is retained only where commissioning confirms a positive separation
Gas and humidity	0.60	0.333	0.10	Volatile solvent evaporating from a spill raised both
Flame and illuminance, emission	0.48	0.267	0.08	Visible-band emission with combustion; optical confirmation of flame, using the emission deviation
Gas and illuminance, obscuration	0.30	0.167	0.05	Attenuation by dense vapour. Many solvent vapours are optically transparent at occupational concentrations, so the separation and coefficient are small
Temperature and illuminance, obscuration	0.30	0.167	0.05	Weak separation through radiant sources and obscuration
Flame and humidity	0.30	0.167	0.05	Combustion products and steam altered local humidity
Humidity and illuminance, obscuration	0.18	0.100	0.03	Condensation on optical surfaces

Table 8: Commissioning Separations, Derived Interaction Coefficients and Associated Mechanism

The mechanisms named above are the scenarios under which each separation was measured. They are candidate interactions confirmed for this installation by commissioning and are not asserted as universal physical laws; a different room or a different reagent set would be expected to produce different separations and therefore different coefficients.

A worked evaluation makes the arithmetic concrete. The condition is 170 ppm isopropanol-equivalent, 33.0 °C, a flame duty fraction of 0.40, 52 %RH and 305 lux, against baselines of 82 ppm, 28.4 °C, zero, 50 %RH and 310 lux with dispersion values of 49.8 ppm, 3.5 °C, 0.70, 10.1 %RH and 22 lux. Every reading lies below its corresponding reference-configuration warning value and every normalized deviation lies below the 2 sigma adaptive warning boundary. Illuminance is below its baseline, so the obscuration deviation is 0.23 and the emission deviation is zero, which sets the flame-light term to zero in this instance.

Step	Quantity	Evaluation	Result
1	n_gas	(170 - 82) / 49.8	1.77
2	n_temp	(33.0 - 28.4) / 3.5	1.31
3	n_flame	(0.40 - 0) / 0.70	0.57
4	n_hum	(52 - 50) / 10.1	0.20
5	n_light, obscuration	(310 - 305) / 22	0.23
6	n_light, emission	max(0, (305 - 310) / 22)	0.00
7	Weighted individual terms	0.35(1.767), 0.25(1.314), 0.25(0.571), 0.07(0.198), 0.08(0.227)	0.618, 0.329, 0.143, 0.014, 0.018
8	Sum of individual terms	0.618 + 0.329 + 0.143 + 0.014 + 0.018	1.122
9	c x n_gas x n_temp	0.30 x 1.767 x 1.314	0.697
10	c x n_gas x n_flame	0.25 x 1.767 x 0.571	0.252
11	c x n_temp x n_flame	0.20 x 1.314 x 0.571	0.150
12	c x n_temp x n_hum and c x n_gas x n_hum	0.15 x 1.314 x 0.198, then 0.10 x 1.767 x 0.198	0.039, 0.035
13	c x n_flame x n_light, emission	0.08 x 0.571 x 0.000	0.000
14	Remaining four pairs	Gas-light, temp-light, flame-humidity, humidity-light	0.042
15	Sum of pairwise interaction terms	0.697 + 0.252 + 0.150 + 0.039 + 0.035 + 0.000 + 0.042	1.215
16	CHI	1.122 + 1.215	2.34

Table 9: Worked Evaluation of the Index for a Sub-Threshold Compound Condition

Two observations follow. The interaction terms contribute 1.215 of the total 2.34, marginally more than the individual terms. And the individual sum of 1.122 alone would place the condition barely inside the warning band, whereas the complete index places it firmly there at 2.34. Detection here is carried by a quantity that no single-channel comparison can produce.

7.5 Decision and Arbitration Layer (302 to 304)

Three decision paths run in parallel and are combined by the arbiter (304), which adopts the highest band called for by any of them.

The first path is the composite index. The band classifier (302) maps CHI to safe below 1.0, warning from 1.0 to below 3.0, and critical at 3.0 and above.

The second path is the per-channel normalized deviation. The same classifier evaluates the designated per-channel standardized deviation of each sensing channel independently against 2 sigma for warning and 3.5 sigma for critical. Each channel designates exactly one such deviation; the emission deviation of the photometric channel is not designated and enters the arrangement only as an operand of the flame pair. This path is a genuine actuator-triggering route, not an analytical convenience, and it is the route by which the escalation records of Section 8.4 reach the warning band. It preserves single-parameter sensitivity in a form that moves with the room, which neither the index nor the absolute comparators provide.

The third path is the absolute backstop comparator (303), which tests selected raw readings against limits written into firmware and never modified by the adaptive stage: 400 ppm isopropanol-equivalent, 45 degrees Celsius, and sustained flame. Not every channel carries an absolute limit. The remaining fixed values of Table 11, being 35 degrees Celsius, 200 ppm, 75 per cent relative humidity and 300 lux, are reference-configuration thresholds used for the comparison of Section 8 and for demonstrating that the compound scenarios of Table 13 lie below every individual warning level; they are neither backstops nor actuator-triggering paths of the invention. The illuminance channel has no absolute limit at all, its hazard direction being a fall rather than a rise.

The arrangement allows adaptive sensitivity and absolute safety to coexist. A purely adaptive system loses its absolute floor, because a baseline that has drifted upward over hours makes a dangerous absolute reading appear ordinary. A purely fixed system loses compound detection and drifts into nuisance activation. An index-only system loses the single-parameter case, as scenarios S3 and S4 show, where the index reaches only 1.72 and 0.58 for conditions that are plainly critical.

Band	Trigger condition	Motor (401)	Servo (402)	Buzzer (403)	Indicator (404)	Activation path
Safe	CHI below 1.0, every designated n_i below 2 sigma, and no reading subject to a limit at that limit	Off	0 degrees	Silent	Green	Monitoring only
Warning	CHI 1.0 to 2.99	50 % duty	45 degrees	1 Hz pulse	Amber	Index path
Warning	Any designated n_i at or above 2 sigma	50 % duty	45 degrees	1 Hz pulse	Amber	Per-channel path
Critical	CHI at or above 3.0	100 % duty	90 degrees	Continuous	Red	Index path
Critical	Any designated n_i at or above 3.5 sigma	100 % duty	90 degrees	Continuous	Red	Per-channel path
Critical	Any raw reading at or above its absolute limit	100 % duty	90 degrees	Continuous	Red	Backstop path

Table 10: Graduated Response Protocol and Activation Paths

7.6 Actuation Layer (400)

The ventilation motor is driven through an H-bridge at three duty levels and the servo sets the exhaust aperture to match. Partial ventilation at the warning stage is the practical benefit of holding a magnitude rather than a flag, since a developing condition receives mitigation graduated to the

adopted band and in most recorded warning events returns to the safe band without reaching the critical stage. The audible and visual outputs distinguish the bands so that an occupant can tell an advisory from an emergency without consulting the interface.

7.7 Communication, Storage and Interface Layer (500)

A packet is assembled every two seconds carrying the five raw readings, the five baselines, the five dispersion values, the five normalized deviations, the index, its individual and interaction components, the band and the actuator state. Packets are pushed over Wi-Fi to a persistent store and rendered by a browser interface refreshing at the same interval. Figure 4 shows the interface. A local display presents current status and remains useful if the wireless link is unavailable.

Recording the component breakdown rather than the index alone is deliberate: an investigator can determine which pair drove a past actuation, which a scalar alone would not support.

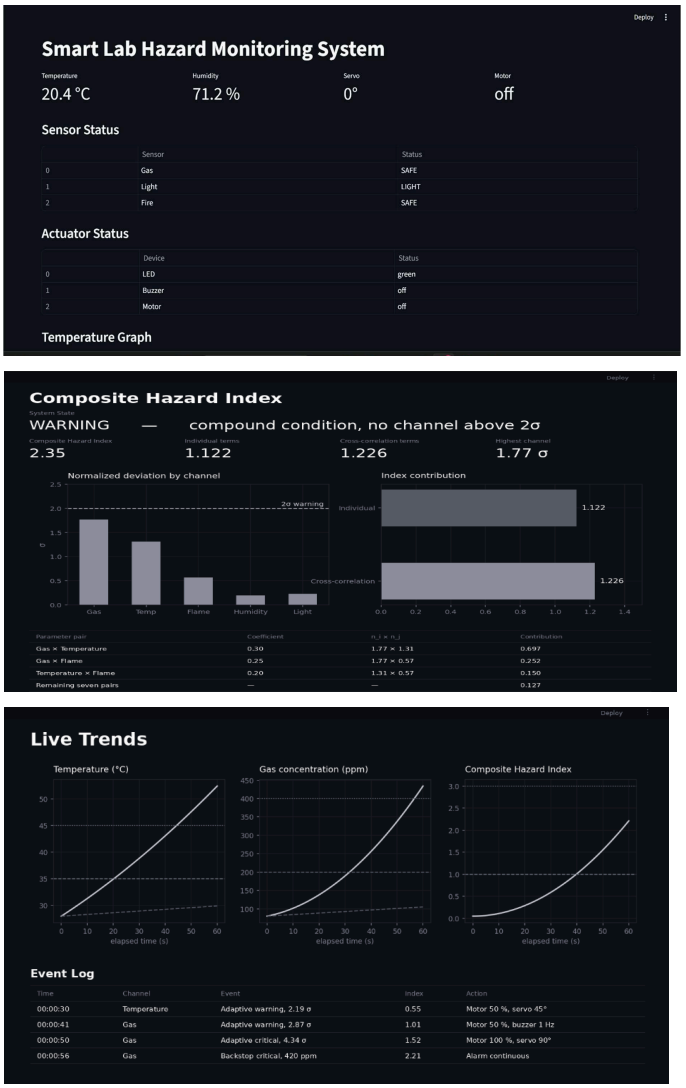


Figure 4: Live Interface Presenting Readings, Sensor States and Actuator States

7.8 Fail-Safe and Constraint Logic

Absolute comparators remain active throughout the ten-minute initialization period, so fixed-limit protection is available from power-up, although compound detection is not available until the registers are populated. A dispersion floor prevents a quiet room producing a near-zero denominator. Deviations in the non-hazard direction are clamped to zero, and because at most one photometric deviation is non-zero at a time the two photometric paths cannot both contribute in one cycle. Loss of a sensing channel is treated as a fault rather than as an absence of hazard: where a channel fails to update within three expected intervals, or returns a value outside its physical range, the unit raises a latched sensor-fault state, indicated on the display and by a distinct amber flash of the indicator (404), recorded with the identity of the failed channel and transmitted in every packet. That channel's

standardized deviation is set to zero and its four interaction terms removed so the index remains computable, and loss of the gas or flame channel forces the ventilation motor to intermediate duty and the servo to 45 degrees until cleared.

7.9 Equivalent Implementations

The described transducers are convenient rather than necessary, and two of them limit the present embodiment. A non-dispersive infrared element would replace the metal-oxide gas sensor with species selectivity and slower drift, removing the need to express concentration as an equivalent against a calibration analyte. A DHT22, SHT31 or thermistor would lift the thermal and hygrometric channels from 0.5 Hz to the full cycle rate, removing the held-value asymmetry of Section 7.2 and cutting the thermal response delay of Table 14 by about 2.25 s. A thermopile array would replace the flame sensor. Any controller offering five inputs, a few hundred kilobytes of memory and a network interface can host the computation. Transport may be Wi-Fi, wired Ethernet or a low-power radio, and the store may be any embedded database. Additional channels extend the index naturally, with interaction terms rising as n(n-1)/2.

7.10 Complete Operating Sequence

Figure 5 sets out the allocation of one control cycle across the eight processing stages. Acquisition and packet assembly with transmission dominate the budget at 90 ms and 80 ms, while fusion takes 48 ms in the worst case and 100 ms remains idle for the local display and for margin. Fusion therefore consumes 9.6 % of the cycle, which is the measured headroom that allows the method to run on inexpensive hardware.

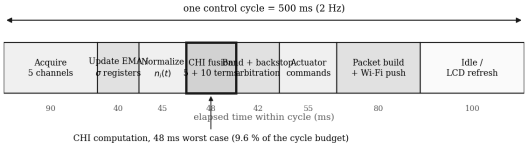


Figure 5: Allocation of One 500 ms Control Cycle Across Processing Stages

7.11 Best Method of Performing the Invention

The best method of performing the invention known to the inventors at the date of this disclosure is the arrangement of Sections 7.1 to 7.10, namely: the five sensing channels of Table 6; the normalization stage of Equations 1 to 3 with the smoothing factor, baseline epoch and dispersion window of Table 7, with dispersion held from the one-sigma excursion entry and baseline held while any actuator is commanded; the fusion module of Equation 4 with the weight vector of Table 7 and the ten coefficients of Table 8, established by the commissioning rule of Equation 5 in the room to be protected; the band classifier with index boundaries at 1.0 and 3.0 and per-channel boundaries at two and three-and-a-half sigma; absolute comparators at 400 ppm isopropanol-equivalent, 45 degrees Celsius and sustained flame, the reference-configuration figures of Table 11 forming no part of the actuation paths of the invention; the three-path arbiter; and the hardware operated at the three levels of Table 10.

8. Experimental Validation Results

8.1 Experimental Setup

Validation was carried out on the prototype of Figure 2, installed on a laboratory bench in an enclosed room of approximately 42 cubic metres with the room ventilation set to its normal working state. Gas, flame and illuminance were sampled at 2 Hz and the DHT11 channels at 0.5 Hz with values held between updates, the index being evaluated at 2 Hz throughout. Baselines, dispersion values, normalized deviations, index values, component breakdowns and actuator states were logged for every cycle.

Four families of the test were executed. Individual-parameter escalation applied a controlled rise to one channel at a time. Compound scenario testing applied combinations chosen so every channel stayed below its individual boundary. A four-hour drift run examined behaviour under slow ambient change and a six-hour run examined stability, communication and resource use. A sixty-event set

spanning compound, single-parameter and benign conditions supported the detection comparison, with a fixed-threshold configuration evaluated on the same recorded events. The interaction coefficients were commissioned in the same room from a staged set that overlaps this evaluation set, so the reported figures characterise the commissioned installation and do not constitute an independent held-out validation.

Measurement error was established against reference instruments, six matched points per channel: a calibrated thermometer, a hygrometer, a reference gas monitor charged with isopropanol vapour, and a lux meter. The gas channel was calibrated against that analyte after a 48 h preheat and all concentrations are reported as ppm isopropanol-equivalent. Table 11 states the thresholds and operating parameters applied throughout. Bench tests share one dispersion set taken from the pre-event steady window: 49.8 ppm for gas, 3.5 °C for temperature, 0.70 for the flame duty fraction, 10.1 %RH for humidity and 22 lux for illuminance. The drift run was conducted under quieter conditions and carries its own value of 0.78 °C.

8.2 Threshold and Operating Parameter Configuration

Parameter	Setting	Type	Condition indicated
Control cycle	2 Hz, 500 ms	Operating	Decision rate
Channel update rates	2 Hz gas, flame, light; 0.5 Hz temperature, humidity	Operating	Transducer-limited acquisition
Baseline update	10 s epoch, a = 0.05; mean of 20 samples at 2 Hz and 5 at 0.5 Hz	Operating	Reference adaptation
Dispersion window	300 s, 600 or 150 samples by channel	Operating	Reference variability
Temperature	35 °C reference, B + 2 sigma adaptive	Warning	Elevated bench temperature
Temperature	45 °C backstop, B + 3.5 sigma adaptive	Critical	Thermal hazard
Gas, ppm isopropanol-equivalent	200 ppm reference, B + 2 sigma adaptive	Warning	Vapour accumulating
Gas, ppm isopropanol-equivalent	400 ppm backstop, B + 3.5 sigma adaptive	Critical	Hazardous concentration for the calibration analyte
Relative humidity	75 %RH reference, B + 2 sigma adaptive	Warning	Evaporation in progress
Illuminance	Below 300 lux reference, B - 2 sigma adaptive	Warning	Obscuration
Flame duty fraction	Sustained at 1.00	Critical	Established flame
Composite index	1.0 and above	Warning	Compound condition developing
Composite index	3.0 and above	Critical	Compound condition confirmed

Table 11: Threshold and Operating Parameter Configuration

Role of the fixed values in Table 11. Table 11 lists two kinds of fixed value and they must not be confused. The three absolute limits of 400 ppm isopropanol-equivalent, 45 degrees Celsius and sustained flame are backstop limits tested by comparator 303, and are an actuator-triggering path of the invention. The remaining fixed values, being 35 degrees Celsius and 200 ppm for warning, 75 per cent relative humidity and illuminance below 300 lux, are not actuator-triggering paths. They serve two purposes only: they constitute the fixed-threshold reference configuration against which the invention is compared throughout Section 8 on the same recorded readings, and they define the individual warning limits below which the compound scenarios of Table 13 are shown to remain. In the invention a warning is asserted only by the composite index reaching 1.0 or by a designated per-channel standardized deviation reaching two sigma.

8.3 Experimental Dataset and Test Scenarios

Table 12 gives the escalation records. Baselines shown are the values held by register 201 at each instant, and normalized deviations follow from those baselines and the dispersion set of Section 8.1.

Test	Time (s)	Reading	Baseline	n (sigma)	Index	Activation path
Thermal	0	28.0 °C	28.00	0.00	0.00	Safe
Thermal	10	30.0 °C	28.00	0.57	0.14	Safe

Test	Time (s)	Reading	Baseline	n (sigma)	Index	Activation path
Thermal	20	33.0 °C	28.10	1.40	0.35	Safe
Thermal	30	36.0 °C	28.34	2.19	0.55	Adaptive warning at 2 sigma
Thermal	40	39.0 °C	28.73	2.93	0.73	Adaptive warning
Thermal	50	43.0 °C	29.24	3.93	0.98	Adaptive critical at 3.5 sigma
Thermal	60	47.0 °C	29.93	4.88	1.22	Backstop critical above 45 °C
Gas	0	80 ppm	80.00	0.00	0.00	Safe
Gas	10	120 ppm	80.00	0.80	0.28	Safe
Gas	20	180 ppm	82.00	1.97	0.69	Safe
Gas	30	230 ppm	86.90	2.87	1.01	Adaptive warning at 2 sigma
Gas	40	310 ppm	94.05	4.34	1.52	Adaptive critical at 3.5 sigma
Gas	50	420 ppm	104.85	6.33	2.21	Backstop critical above 400 ppm

Table 12: Individual-Parameter Escalation Records

Table 13 gives the seven scenarios used for the compound comparison. Scenarios S1 to S4 are single-parameter or benign conditions included as controls. Scenarios S5 to S7 are the compound conditions, constructed so that every channel remains below both the corresponding reference-configuration warning value and its 2 sigma adaptive boundary.

Ref	Scenario	Gas (ppm)	Temp (°C)	Flame	Individual sum	Index	Outcome comparison
S1	Normal operation	88	28.9	0.00	0.11	0.13	Both correctly silent
S2	Gas warning, single	220	28.9	0.00	1.04	1.25	Both warn
S3	Thermal critical, single	88	46.0	0.00	1.33	1.72	Both critical, backstop path
S4	Flame present, single	88	28.9	1.00	0.47	0.58	Both critical, backstop path
S5	Compound, sub-threshold	170	33.0	0.40	1.12	2.34	Fixed silent, index warns
S6	Compound, elevated	180	34.5	0.40	1.30	2.95	Fixed silent, index warns
S7	Compound, early onset	150	31.0	0.35	0.82	1.45	Fixed silent, index warns

Table 13: Compound and Control Scenario Matrix

8.4 Individual-Parameter Escalation Analysis

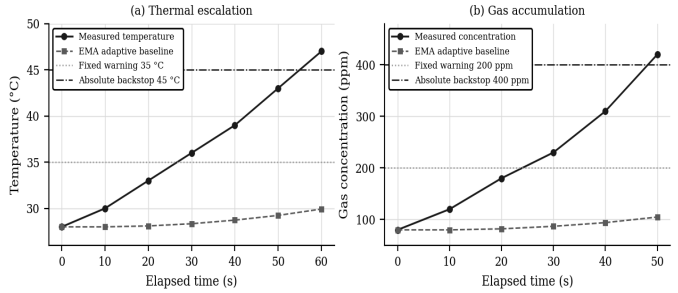


Figure 6: Thermal and Gas Escalation Against Adaptive Baselines and Absolute Limits

Figure 6 plots both escalation records with the adaptive baseline and the two absolute limits. In the thermal record the baseline rises from 28.00 °C to only 29.93 °C while the reading rises to 47.0 °C, which is the intended behaviour: a 19 °C excursion completed in one minute is absorbed by the baseline to the extent of 1.93 °C. The normalized deviation therefore grows monotonically to 4.88 sigma rather than flattening.

Protection escalates through two of the three paths in sequence. The per-channel path reaches its 2 sigma warning boundary at 30 s and its 3.5 sigma critical boundary at 50 s, when the reading is 43.0 °C and still 2 °C below the absolute limit. The backstop path asserts at 60 s. The per-channel adaptive path therefore leads the absolute path by 30 s in this record, and the absolute path confirms rather than initiates. Note that the thermal channel updates at 0.5 Hz, so the times quoted carry a quantization of up to 2 s.

The gas record behaves the same way, reaching the adaptive warning at 30 s and the backstop at 50 s. The index remains low throughout both records, at 1.22 and 2.21 at the final points, because a single elevated channel generates one weighted term and no product of consequence. This is the expected signature of a single-parameter event, and it is the reason the backstop path exists.

8.5 Compound Hazard Detection Analysis

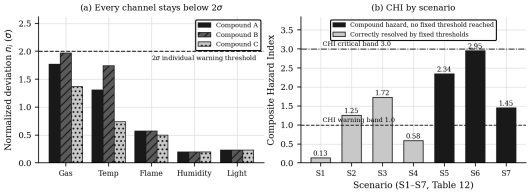


Figure 7: Normalized Deviations and Index Values Across the Scenario Set

Panel (a) of Figure 7 plots the normalized deviations for the three compound scenarios against the 2 sigma boundary. The largest value anywhere is 1.97 sigma, on the gas channel of S6. No channel reaches its individual warning boundary, and no raw reading reaches either its absolute limit or the corresponding reference-configuration warning value, the highest being 180 ppm against the 200 ppm reference value and 34.5 °C against the 35 °C reference value.

Panel (b) plots the index for all seven scenarios. The three compound conditions return 2.34, 2.95 and 1.45, all inside the warning band. The fixed-threshold configuration returns no output for any of the three. For the four control scenarios the two configurations agree, which confirms that compound sensitivity has not been obtained by lowering the effective sensitivity elsewhere.

The reason is visible in the arithmetic of Table 9. Three moderate deviations that sum to a modest weighted total produce products that are individually small but numerous, and the sum of those products is comparable to the weighted sum itself. The response boundary is therefore crossed by a term that cannot exist in any arrangement lacking either the standardization stage or the pairwise products. Because the coefficients were commissioned against a staged set overlapping these scenarios, the margin reported here should be read as characterising the commissioned installation.

8.6 Index Component Contribution and Ablation Analysis

Separating each index value into its weighted individual component and its interaction component amounts to an ablation of the inventive step, since what remains is an ordinary weighted sum of standardized deviations.

For S5 the split is 1.122 individual against 1.215 interaction. For S6 it is 1.30 against 1.65, and for S7, 0.82 against 0.63. In the two stronger compound cases the interaction terms carry the majority of the magnitude. For S7 the individual sum alone is 0.82, which sits below the warning boundary of 1.0, so the ablated configuration would produce no warning at all. The complete index returns 1.45 and warns.

The control scenarios behave in the opposite way. For S2, a single elevated gas channel, the interaction component is 0.21 against an individual component of 1.04, and for S1 it falls to 0.02, because no second elevated operand exists for the products to act on. The interaction machinery is inert in ordinary and single-parameter conditions and active only where related channels move together, which is the selectivity the design requires.

8.7 Response Delay and Latency Decomposition

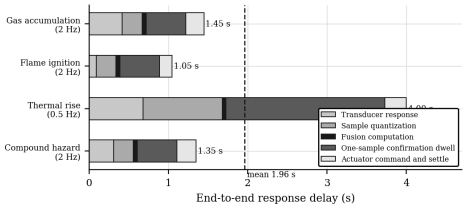


Figure 8: Response Delay Decomposition by Event Type

Event	Channel rate	Transducer response (s)	Quantization (s)	Fusion (s)	Confirmation dwell (s)	Actuator settle (s)	Total (s)
Gas accumulation	2 Hz	0.42	0.25	0.048	0.50	0.23	1.45
Flame ignition	2 Hz	0.09	0.25	0.048	0.50	0.16	1.05

Event	Channel rate	Transducer response (s)	Quantization (s)	Fusion (s)	Confirmation dwell (s)	Actuator settle (s)	Total (s)
Thermal rise	0.5 Hz	0.68	1.00	0.048	2.00	0.27	4.00
Compound hazard	2 Hz	0.31	0.25	0.048	0.50	0.24	1.35

Table 14: End-to-End Response Delay Decomposition

Quantization is taken as half the update interval of the triggering channel and the confirmation dwell as one further sample of that channel, so both scale with the channel rate rather than with the control cycle. The mean of the four totals is 1.96 s.

The decomposition locates the cost. Transducer response varies by nearly an order of magnitude, from 0.09 s on the flame channel to 0.68 s on the thermal channel. The dominant term for the thermal event is not the transducer but the 0.5 Hz update rate of the DHT11, which contributes 3.00 s of the 4.00 s total through quantization and dwell. A transducer capable of the full 2 Hz cycle would reduce that event to about 1.75 s and the four-event mean to about 1.40 s, which identifies the component change with the largest effect on responsiveness.

Fusion contributes 48 ms in every case, 2.4 % of the mean delay, and is therefore not a limiting factor. The compound event completes in 1.35 s, faster than the gas event, which matters because that path has no fixed-threshold alternative: a condition that would otherwise be missed entirely is detected at no latency penalty.

8.8 Detection Accuracy and Measurement Error Analysis

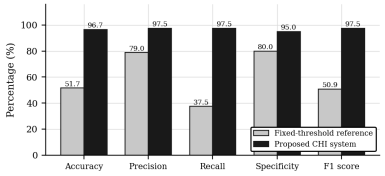


Figure 9: Detection Performance, Proposed Index Against a Fixed-Threshold Reference

The sixty-event set comprised 24 compound conditions, 16 single-parameter conditions and 20 benign disturbances such as door movement, solvent wiping, instrument warm-up and direct sunlight on the bench. Both configurations were evaluated on the same recorded events, and Figure 9 plots the outcome.

Configuration	TP	FN	TN	FP	Derived metrics
Proposed index system	39	1	19	1	Accuracy 96.67 %, precision 97.50 %, recall 97.50 %, specificity 95.00 %, F1 97.50 %
Fixed-threshold reference	15	25	16	4	Accuracy 51.67 %, precision 78.95 %, recall 37.50 %, specificity 80.00 %, F1 50.85 %

Table 15: Detection Confusion Matrices and Derived Metrics

Metrics follow from the matrices by accuracy = (TP + TN) / n, precision = TP / (TP + FP), recall = TP / (TP + FN), specificity = TN / (TN + FP) and F1 = 2PR / (P + R). Accuracy improves by 45.00 percentage points, equivalent to 87.1 % in relative terms. Recall improves by 160.0 % relative, which is the dominant effect and reflects the 24 compound events the fixed configuration cannot reach. False positives fall from four to one, a reduction of 75.0 %, attributable to baseline adaptation rather than to the fusion stage.

Channel	MAE	RMSE	MAPE (%)	Effect on the index
Temperature	1.53 °C	1.55 °C	4.52	0.44 sigma at the bench dispersion of 3.5 °C
Relative humidity	4.05 %RH	4.06 %RH	6.69	0.40 sigma at 10.1 %RH
Gas, isopropanol-equivalent	16.33 ppm	17.99 ppm	5.75	0.33 sigma at 49.8 ppm
Illuminance	26.00 lux	29.81 lux	5.63	1.18 sigma at 22 lux
Flame, binary decision	Accuracy 95.00 %	Precision 96.55 %	Recall 93.33 %	F1 94.92 % over 60 trials

Table 16: Measurement Error Against Reference Instruments

Table 16 gives the measured error, which is consistent with the manufacturer specifications of the selected transducers and is not negligible relative to channel dispersion. Temperature error at 1.53 °C is 0.44 sigma and humidity error at 4.05 %RH is 0.40 sigma, so a compound condition near a band boundary could be displaced across it by transducer error alone. Illuminance is worse at 1.18 sigma. That channel carries the smallest weight and its largest coefficient is 0.08, so its contribution stays

bounded, accounting for 0.018 of an index of 2.34 in S5. These magnitudes follow directly from the low-cost transducers chosen and are the strongest argument for the substitutions of Section 7.9. They do not affect the comparison in Table 15, since both configurations used the same recorded readings.

8.9 Adaptive Baseline Drift and False-Warning Analysis

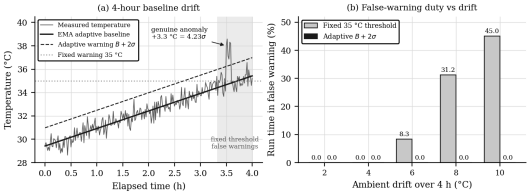


Figure 10: Baseline Tracking Under Drift and False-Warning Duty Against Drift Magnitude

The four-hour run applied 6.0 °C of ambient drift from a starting condition of 29.5 °C, a rate of 1.50 °C per hour, with a genuine anomaly injected at 3.5 hours. Panel (a) of Figure 10 shows the record.

The baseline follows the ramp with a steady-state tracking error of 0.079 °C, from the ramp rate of 0.00417 °C per epoch and the factor (1 - a) / a = 19. That is 0.101 sigma at the drift-run dispersion of 0.78 °C, so drift alone moves the deviation by about a tenth of a sigma against a boundary of two. The adaptive paths recorded no false warning.

The fixed 35 °C threshold was passed at 3.67 hours and remained passed for the rest of the run, holding a false warning state for 8.3 % of the four hours, which is 20.0 minutes. Panel (b) extends this to five drift magnitudes.

Drift over 4 h (°C)	Peak reading (°C)	Fixed threshold duty (%)	Adaptive duty (%)	Fixed threshold time in false warning
2.0	31.5	0.0	0.0	None
4.0	33.5	0.0	0.0	None
6.0	35.5	8.3	0.0	20.0 min
8.0	37.5	31.2	0.0	75.0 min
10.0	39.5	45.0	0.0	108.0 min

Table 17: False-Warning Duty and Baseline Response Under Drift

The adaptive boundary recorded no false warning at any magnitude tested, because it is referenced to a quantity that moves with the room. The genuine anomaly of 3.3 °C above baseline was detected as a 4.23 sigma excursion, above the 3.5 sigma critical boundary, at an absolute reading of 38.0 °C that the fixed 35 °C threshold would have reported with the same weight as the preceding two hours of drift. Sensitivity to a real event was preserved while nuisance activation was removed.

Baseline recovery after a step change follows 0.95 raised to the number of epochs. Reaching 95 % of a step requires 58.4 epochs, which is 9.7 minutes, and 99 % requires 15.0 minutes. Recalibration after a change of operating condition therefore completes without operator action.

8.10 Ventilation Effectiveness and Continuous-Operation Results

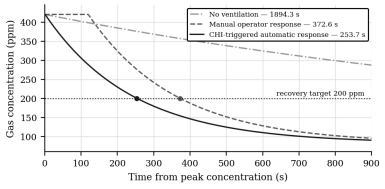


Figure 11: Gas Concentration Recovery Under Three Response Conditions

Recovery from a peak of 420 ppm toward the 200 ppm target was recorded under three conditions, with an ambient floor of 82 ppm. Decay follows C(t) = C_amb + (C_0 - C_amb) exp(-(t - t_d) / tau), so the time to target is t_d + tau ln(338 / 118) = t_d + 1.0524 tau.

Response condition	tau (s)	Dead time (s)	Time to target (s)	Time to target (min)
No ventilation	1800	0.0	1894.3	31.57
Manual operator response	240	120.0	372.6	6.21
Index-triggered automatic response	240	1.1	253.7	4.23

Table 18: Gas Concentration Recovery to the 200 ppm Target

Figure 11 shows automatic response reaching the target 118.9 s sooner than manual response, an improvement of 31.9 %, and 1640.6 s sooner than no response, an improvement of 86.6 %. The two

ventilated conditions share the same time constant, so the entire advantage comes from removing the 120 s of human reaction and travel time.

Metric	Measured	Requirement	Outcome
Continuous operation without restart	6.0 h	4 h or more	Met
Control cycles executed	43 200	Not applicable	Met
Packets transmitted and acknowledged	10 800 sent, 10 477 acknowledged	Not applicable	Met
Packet delivery success and loss	97.00 % success, 2.99 % loss	95 % or more, loss below 5 %	Met
Interface latency, mean and worst	0.68 s and 0.74 s	Below 3 s	Met
Fusion computation, mean and worst	41.6 ms and 48 ms, 9.6 % of cycle	Below 100 ms	Met
Heap occupancy	218 of 320 kB, 68.1 %	Below 85 %	Met
Compound detections in the 24-event subset	24 of 24, 100 %	90 % or more	Met
False warnings during the run	0	Below 3	Met

Table 19: Continuous-Operation Results Over Six Hours

Hourly delivery success ranged from 96.5 % to 97.4 % with a mean of 97.00 %, and latency from 0.62 s to 0.74 s with a mean of 0.682 s. Neither quantity trended across the six hours, so neither memory growth nor thermal effects degraded the link.

8.11 Overall Performance Metrics and Derived Measurements

Metric	Proposed system	Reference configuration	Source
Accuracy, F1, recall, specificity	96.67, 97.50, 97.50, 95.00 %	51.67, 50.85, 37.50, 80.00 %	Table 15
False positives in 60 events	1	4	Table 15
Compound detection rate	24 of 24, 100 %	0 of 24, 0 %	Tables 12, 18
Highest deviation at detection	1.97 sigma against a 2 sigma boundary	Not applicable	Table 13
Interaction share of index, S5	52.0 %, 1.215 of 2.337	Not applicable	Table 9
Mean response delay, four events	1.96 s	Not applicable	Table 14
Worst fusion time, heap use	48 ms, 9.6 % of cycle; 68.1 %	Not applicable	Table 19
False-warning duty at 6 °C drift	0.0 %	8.3 %	Table 17
Baseline recovery to 95 %	9.7 min, automatic	Manual recalibration	Section 8.9
Recovery to 200 ppm from 420 ppm	253.7 s	372.6 s manual, equal time constant	Table 18
Packet delivery over 6 h	97.00 %	Not applicable	Table 19

Table 20: Summary of Derived Performance Metrics

8.12 Discussion of Results and Integrated Technical-Effect Summary

Table 20 collects the derived metrics, and every objective stated in Section 5 is supported by one of them. The compound detection objective is met by three staged conditions returning index values of 1.45, 2.34 and 2.95 with no channel above 1.97 sigma. The normalization objective is met by the worked evaluation of Table 9, which combines a concentration, a temperature, a duty fraction, a humidity and an illuminance into one scalar. The drift objective is met by zero adaptive false warnings against 8.3 % duty for the fixed threshold, and the embedded-resource objective by 48 ms of a 500 ms cycle and 68.1 % heap occupancy. The three decision paths are demonstrated separately: the index resolves S5 to S7, which no other path reaches; the absolute comparator forces the critical response in S3 and S4, where the index alone reaches 1.72 and 0.58; and the per-channel path precedes the comparator during escalation, reaching warning at 30 s and critical at 50 s against the comparator's 60 s.

The strongest single result is the recall improvement from 37.50 % to 97.50 %, which comes almost entirely from the 24 compound events. The tested fixed-threshold configuration did not detect any of those events. No claim is made that no fixed-threshold setting whatsoever could detect them, since only the one reference configuration was evaluated.

Three trade-offs are acknowledged. Coefficients are established during commissioning from staged conditions in the target room, which makes the detection behaviour specific to the installation and requires a step a factory-configured detector does not; the compensating benefit is installation-specific interaction coefficients rather than factory defaults. The reported detection figures come from a set

overlapping the commissioning set, so they characterise a commissioned installation rather than generalised performance. The selected transducers impose real limits: the DHT11 update rate dominates the thermal response delay, and measurement error reaches 0.44 and 0.40 sigma on the thermal and hygrometric channels. Table 21 traces each technical effect to its mechanism and evidence.

Technical effect	Mechanism producing it	Quantitative evidence
Detection of hazards below every individual threshold	Pairwise products of standardized deviations	Index 1.45 to 2.95 with all channels below 1.97 sigma, Table 13
Arithmetic combination of dissimilar physical quantities	EMA baseline and dispersion acting as a normalization stage	Five units combined into one scalar, Table 9
Sensitivity retained under environmental drift	Baseline referenced to the installed room	0.079 °C tracking error, 0.101 sigma, Section 8.9
Nuisance activation removed	Adaptive boundary moves with the baseline	0.0 % against 8.3 % duty at 6 °C drift, Table 17
Overall detection improvement	Fusion and adaptation acting together	Accuracy 51.67 % to 96.67 %, F1 50.85 % to 97.50 %, Table 15
Absolute protection preserved	Backstop comparator arbitrated by logical OR	S3 and S4 critical via backstop while index below 2.0, Table 13
Graduated early mitigation	Continuous index rather than a binary flag	Warning band at 50 % duty and 45 degrees, Table 10
Faster clearance of an accumulated hazard	Automatic actuation removes human reaction time	253.7 s against 372.6 s, 31.9 % faster, Table 18
Execution within a low-cost controller	Ten pair products plus fifteen weighted accumulations per cycle	48 ms of 500 ms, 9.6 %, heap 68.1 %, Table 19

Table 21: Technical Effect, Mechanism and Quantitative Evidence

8.13 Key Validation Insights, Limitations and Future Scope

The pairwise interaction terms supplied 52.0 % of the index in scenario S5, being 1.215 of 2.337, and were decisive in S7, where the individual sum of 0.82 alone would have produced no warning against a boundary of 1.0.

Compound sensitivity was obtained without loss of selectivity: the interaction component fell to 0.02 under normal operation and 0.21 for a single elevated channel, and specificity across the set was 95.00 %.

The two mechanisms address different failure modes and are separable in the evidence: fusion produced the recall gain from 37.50 % to 97.50 %, while adaptation produced the false-positive reduction and eliminated drift-induced nuisance warnings. The per-channel adaptive path led the absolute path by 30 s in the thermal escalation record, so the absolute path confirms rather than initiates, and the 48 ms cost of fusion does not appear in a mean end-to-end delay of 1.96 s dominated by transducer response.

Several limitations are recognized. Coefficients are fixed after commissioning, so a substantial change in room usage would call for re-commissioning. The metal-oxide gas element is not species-selective and shows the drift documented in the reviewed literature, which the adaptive baseline mitigates without removing. Validation covered six hours of continuous operation and a four-hour drift run, so long-term ageing lies outside the present evidence. The index reports severity rather than identity: the component breakdown indicates which pair drove the response but does not name the species involved.

Development would proceed along five lines: an independent held-out event set commissioned separately from the evaluation set; a non-dispersive infrared element for species selectivity; a faster hygrothermal transducer, removing the 0.5 Hz constraint and cutting the mean response delay to about 1.40 s; online refinement of coefficients from accumulated operating records; and closed-loop control driven by the rate of change of the index.

9. Discussion

9.1 Achievement of the Stated Objectives

The evidence of Section 8 supports each objective with a measurement. Compound conditions reaching no individual threshold were detected in 24 of 24 instances, with the highest normalized deviation in that set at 1.97 sigma against a boundary of 2. Detection accuracy reached 96.67 % against 51.67 % for a fixed-threshold configuration on the same recordings, subject to the

commissioning overlap noted in Section 8.1. Nuisance activation under drift was eliminated. Response completed in a mean of 1.96 s, and fusion consumed 9.6 % of the control cycle.

9.2 Inventive Step of the Claimed Combination

The objection to be met is that every operation in this invention is individually known, and Section 3 concedes exactly that. Products of baseline-referenced sensor signals were published in 1992. Standardization against a running mean and dispersion appears in US5079422A, US20190108739A1 and US7202794B2. Weighted combination of detector outputs with a retained individual path appears in US5691703A, parallel operation of a statistical and a correlation path in US4665390A, and fan actuation from a gas threshold in CN102645908A. The chain from heterogeneous measurement through common normalization and composite scoring to automatic ventilation appears in US20220316730A1; a standardized temperature-by-humidity interaction term gating an actuator appears in CN121264336A; and a sum of pairwise products added to a weighted individual sum appears in CN122217398A. No assertion is made that any of those operations is itself new, and the case for inventive step rests on the cooperative arrangement rather than on any element of it.

Three arguments are advanced.

The first concerns commensurability. Heskestad and Newman multiplied baseline-referenced differences that retained their physical units, and reported two hand-selected pairings; products of that form can be scaled by dimensioned coefficients, so no assertion is made that combination is impossible. What is not disclosed there is the standardization of each channel by its own running dispersion to yield a common dimensionless representation, nor the summation of a plurality of such standardized pairwise terms together with standardized individual terms in one expression. The prior standardization art performs that division but does not multiply its outputs together; the prior product art multiplies but does not standardize. US20220316730A1 normalizes to a common scale but combines by weighted average, which cannot exceed the range of its own operands. CN121264336A multiplies two standardized factors but carries one interaction term only, inside a multiplicative risk expression.

The second concerns protection of the reference, and the act of inhibition is conceded to be old: WO2021257287A1 freezes a baseline during a detected interaction and the Eaton reference excludes event-related values from an adaptive threshold buffer. An adaptive reference has a failure mode that must be closed: as an event develops, statistics computed from the event widen the tolerance band against which the event is judged, and once mitigation begins the actuator output is learned as the new normal. Holding the dispersion register from the moment any channel passes one sigma, and holding the baseline register while mitigation runs, closes both routes. Event-triggered inhibition of a statistic used for threshold estimation is known in the anomaly-detection and gas-detector arts, and no novelty is asserted in the act of freezing a statistic as such. What is relied upon is that the protected statistics are the operands of heterogeneous pairwise interaction terms, and that the same protection therefore determines what those products are able to express, in combination with the three-path arbitration and the mitigation feedback of Section 7.8.

The third concerns the decision structure. US4665390A places two paths in parallel, both operating on radiation channels of the same modality. Here three paths operate on different descriptions of the same environment: a composite magnitude across all channels, per-channel bands that move with the room, and fixed absolute limits. It is not asserted that two paths would leave any scenario of Table 13 unresolved. The per-channel adaptive path is retained because it acts earlier and against a reference that moves with the room: in the thermal escalation record it reaches warning at 30 s and critical at 50 s against the comparator's 60 s, and in Table 17 it is that path which holds the false-warning duty at zero under drift. The output of the arbitration is a magnitude setting ventilation duty and exhaust aperture, which no identified reference produces.

The ablation of Section 8.6 supports the argument quantitatively. Removing the interaction terms leaves the weakest compound condition at 0.82 against a boundary of 1.0, and removing baseline adaptation restores the drift-induced nuisance activation quantified in Table 17. Neither element alone reproduces the observed characteristic.

9.3 Statutory Character Under Sections 3(k) and 3(f)

Section 3(k) of the Indian Patents Act excludes mathematical methods, business methods, computer programmes per se and algorithms. The relevant question under current examination practice for computer-related inventions is not whether novel hardware is present but whether the invention considered as a whole solves a technical problem by technical means and produces a technical effect beyond the ordinary operation of a program.

The technical problem here is that independent detectors in an enclosed room fail to act on a hazard distributed across several parameters, and that fixed thresholds lose sensitivity as the room drifts. The technical means are physical: heterogeneous transducers, statistics maintained from and protected against the physical state of the installation, and arbitration that commands ventilation duty and exhaust aperture. The technical effect is a change in the physical state of the room: earlier engagement of ventilation hardware, a measured reduction in concentration, and elimination of spurious actuation under ambient drift.

Claims should accordingly be framed as a hazard-detection and control system in which sensor signals are converted to protected standardized deviations, combined, arbitrated and used to operate mitigation hardware. An independent claim directed primarily to the expression for CHI would invite the exclusion and should be avoided. The expression belongs in the claim as the means by which the sensed state is converted into an actuation decision, not as the subject of the claim.

Section 3(f) excludes mere arrangement or re-arrangement of known devices each functioning independently. The parts here do not function independently. The dispersion estimator determines what the fusion module can express; the fusion module determines what the arbiter sees; the arbiter determines the actuator state; and the actuator state feeds back to inhibit the baseline estimator that conditions the whole chain. Section 8.6 demonstrates that removing any one element changes the behaviour of the others. That mutual conditioning, rather than the collection of a sensor, a processor, a fan and an alarm, is what is put forward as the invention.

9.4 Deployment and Residual Limitations

The unit runs on inexpensive hardware, requires no trained classifier and no external machine-learning training process, the interaction coefficients being obtained from a finite installation-specific commissioning set, and needs no external computation, so a room can be protected by a single self-contained device commissioned in place. Extension to further channels follows the same construction, with interaction terms growing as n(n-1)/2, so a sixth channel would add five pairwise terms. On the headroom measured for the five-channel implementation, 48 ms of the 500 ms cycle, that extension is expected to remain within budget, but it has not been built or measured and no figure is asserted for it. The residual limitations are those of Section 8.13.

10. What Aspect(s) of the Invention Need(s) Protection?

Protection is sought for the cooperative arrangement set out below, ordered into three tiers because the individual operations are old and only their interaction is put forward as the invention. Section 3 identifies prior art for each operation taken alone.

10.1 Independent-Claim Core

An embedded hazard detection and mitigation system comprising at least four sensing channels responsive to measurands in distinct physical domains; a normalization stage maintaining for each channel a running-average baseline register and a dispersion register and producing a standardized deviation taken in the hazard direction of that channel, updating of the dispersion register being inhibited for the duration of an excursion and updating of the baseline register while a mitigation actuator is operating; a fusion stage forming a composite magnitude as the sum of weighted individual standardized deviations and weighted pairwise products of selected standardized deviations produced by the normalization stage, at least one such coefficient being non-zero; and an arbitration stage operating the mitigation actuator at a level selected from at least three levels according to the highest band called for by the composite magnitude, by any designated per-channel standardized deviation, or by comparison of a reading against a fixed absolute limit. The exponentially weighted form of the running average, and non-negativity of the weights and coefficients, are preferred implementations

claimed dependently. Both independent claims require the coefficients and bands to be so configured that the composite magnitude can enter a band above the lowest while every designated per-channel deviation remains below its warning boundary.

The features that must appear together in the independent claim, because the effect depends on their interaction rather than on any of them separately, are these.

- Division of each baseline-referenced difference by a per-channel dispersion estimate, so that the products formed downstream are unit-free and mutually comparable. This is what distinguishes the arrangement from the multiplication of dimensioned baseline-referenced signals in the identified literature.
- Formation of pairwise products from standardized deviations produced by the normalization stage, the operands of a product not being restricted to those deviations that carry an individual weight, and summation of the products together with the weighted individual deviations in one expression.
- Inhibition of dispersion updating during an excursion and of baseline updating during mitigation, so that the operands of those products are protected from the event and from the response to it.
- Designation, for each sensing channel, of one standardized deviation as its per-channel deviation for banding purposes, a channel yielding a second deviation being free to use that second deviation as an interaction operand only.
- Arbitration across the composite path, the per-channel path and the absolute path such that no path can suppress another, and selection of an actuator level from three or more levels according to the arbitrated band.

10.2 Dependent-Claim Fallback Features

- Derivation of the pairwise coefficients by the ordered commissioning rule of Section 7.4, in which a separation statistic is formed for each pair from staged hazard and benign events, a scale factor places a designated onset condition at the warning boundary, the coefficients are derived and quantized, and the boundary condition is re-verified within a stated tolerance.
- Production of two standardized deviations from a single photometric transducer, one for falling illuminance carrying the channel weight and one for rising illuminance serving as the operand of the flame pair, whereby one transducer serves both an obscuration hazard and optical confirmation of flame.
- Pairing of a chemical channel with a thermal channel and of a chemical channel with an optical combustion channel, weighted so that joint sub-threshold elevation of either pair produces an actionable magnitude.
- Separation of the baseline update epoch from the channel sampling rate and from the decision rate, whereby the reference tracks drift over hours while remaining insensitive to excursions lasting seconds.
- Evaluation of the composite magnitude at a common decision rate while individual channels update at differing transducer-limited rates, with the most recent value held between updates.
- Setting of the coefficient scale factor such that a designated onset condition evaluates to the warning boundary within the specified tolerance.
- Operation in which the weighted individual component alone remains below the first boundary while addition of the pairwise component carries the composite magnitude across it, as demonstrated by scenario S7 of Section 8.5, where the individual sum of 0.82 lies below the boundary of 1.0 and the complete index reaches 1.45. Recording of the individual and pairwise components separately from the magnitude, whereby the pair responsible for an actuation is identifiable after the event.

10.3 Embodiment Details

The following are disclosed as one worked implementation and are not put forward as independently protectable: the specific transducers and their reference numerals; the parameter values of Table 7; the particular weight vector and the ten coefficient values; the band boundaries at 1.0 and 3.0 and the per-channel boundaries at two and three-and-a-half sigma; the absolute limits of 400 ppm

isopropanol-equivalent, 45 °C and sustained flame; the reference-configuration values of Table 11; and the ESP32 controller, the 2 Hz cycle, the Wi-Fi transport, the SQLite store and the browser interface.

Equally, the following are acknowledged as known and are not claimed standing alone: exponentially weighted moving averages; standardization of a reading against its own dispersion; sensor fusion in general; multiplication of two sensor signals; weighted summation; combination of an alarm path with a backup threshold; three-level alarm signalling; fan speed control; servo-operated vents; and inhibition of an adaptive baseline during the event it is intended to reveal, as in WO2021257287A1.

10.4 Claims

The following claims are presented in the form in which protection is sought. Two independent claims are presented, Claim 1 drawn to the system and Claim 17 to the evaluation and arbitration unit. The tiering of Sections 10.1 to 10.3 indicates which features are regarded as essential and which are dependent fallbacks. Reference numerals do not limit the scope of the claims.

Claim 1 (Independent): Compound Hazard Detection and Mitigation System A hazard detection and mitigation system for an enclosed environment, the system comprising:

(a) at least four sensing channels (101 to 105) responsive to measurands in distinct physical domains, each arranged to produce a reading at a respective channel update rate;

(b) a normalization stage (200) comprising, for each said channel, a baseline register (201) holding a running average of the readings of that channel, a dispersion register (202) holding a dispersion estimate of that channel over a trailing window, and a normalizer (203) arranged to produce for each channel at least one standardized deviation as the positive part of a signed difference between the reading and the baseline, divided by the dispersion estimate, the sign of that difference being taken in the hazard direction of the channel concerned so that a channel whose hazard direction is a fall in the measured quantity yields a standardized deviation from the amount by which the reading falls below the baseline, whereby the standardized deviations of channels of differing physical domain and differing hazard direction are rendered dimensionless and mutually comparable;

(c) a protection gate (205) configured to inhibit updating of the dispersion registers while any said standardized deviation exceeds an excursion entry value, and to inhibit updating of the baseline registers while any mitigation actuator is commanded above its rest state;

(d) a fusion module (301) configured to form a composite hazard index as the sum of weighted individual standardized deviations and weighted pairwise products of selected standardized deviations produced by the normalization stage, at least one coefficient of the said pairwise products being non-zero;

(e) a band classifier (302) configured to assign the composite hazard index to one of at least three bands, and to assign to one of at least three bands, for each said sensing channel, at least one standardized deviation of that channel designated as its per-channel deviation;

(f) an absolute backstop comparator (303) configured to compare one or more of said readings against a respective fixed limit not modifiable by the normalization stage;

(g) an arbiter (304) configured to adopt the highest band called for by the composite hazard index, by any said designated per-channel standardized deviation, or by the absolute backstop comparator, no one of said three paths being capable of suppressing another; and

(h) at least one mitigation actuator (401, 402) operable at three or more levels and an indicating means (403, 404), the actuator being commanded to the level corresponding to the band adopted by the arbiter,

the said coefficients and the said bands of the composite hazard index being configured such that the composite hazard index is capable of entering a band above the lowest band while every said designated per-channel standardized deviation remains below its own warning boundary and no reading that is subject to a fixed limit has reached that limit, the system being thereby capable of mitigating a hazard distributed across a plurality of said channels at a stage at which no single channel would call for any action, and of mitigating a hazard in which an individual channel exceeds its boundary or its fixed limit.

Claim 2: The system of Claim 1, wherein the said at least three bands of the composite hazard index comprise a safe band below a first boundary, a warning band between the first boundary and a second boundary, and a critical band at or above the second boundary, and wherein the mitigation actuator is commanded to a rest level, an intermediate level and a full level respectively.

Claim 3: The system of Claim 2, wherein the mitigation actuator comprises a ventilation motor (401) commanded to zero, an intermediate and a full duty cycle, and a servo actuator (402) commanded to close, partially open and fully open an exhaust aperture, whereby ventilation is applied at a level graduated by the adopted band.

Claim 4: The system of Claim 1, wherein one of said sensing channels is a photometric channel (105) and the normalizer (203) is configured to produce therefrom two standardized deviations, being an obscuration deviation formed from the fall of the reading below the baseline and an emission deviation formed from the rise of the reading above the baseline, the obscuration deviation being the designated per-channel deviation of that channel and carrying its weight in the said sum of individual terms, and the emission deviation being used solely as an operand of a pairwise product formed with an optical combustion channel and not being assigned to a band, whereby a single transducer serves both an obscuration hazard and optical confirmation of flame.

Claim 5: The system of Claim 4, wherein at most one of the obscuration deviation and the emission deviation is non-zero at any instant.

Claim 6: The system of Claim 1, wherein the said channels comprise a chemical channel (101), a thermal channel (102), a hygrometric channel (103), an optical combustion channel (104) and a photometric channel (105), and wherein the pairwise product formed between the chemical channel and the thermal channel carries the largest said coefficient and the pairwise product formed between the chemical channel and the optical combustion channel carries the second largest.

Claim 7: The system of Claim 2, wherein the said coefficients of the pairwise products are determined during commissioning of the system in the environment to be protected by the ordered steps of: forming, for each pair, a separation statistic as the difference between the median of the product of the two standardized deviations observed during staged hazard events and the median of the same product observed during benign disturbance events; determining an unquantized scale factor that places a designated onset condition at the boundary of the warning band; deriving each coefficient as the product of that scale factor and the ratio of its separation statistic to the largest separation statistic; quantizing the derived coefficients to a stated precision; and verifying that the designated onset condition evaluates to the said boundary within a stated tolerance, the scale factor being re-determined and the steps repeated where it does not.

Claim 8: The system of Claim 7, wherein a coefficient whose separation statistic does not exceed a predetermined minimum is set to zero, whereby the corresponding pairwise product is removed from the composite hazard index.

Claim 9: The system of Claim 1, wherein the baseline registers (201) are updated once per update epoch from the mean of the samples acquired within that epoch, the update epoch being longer than the sampling interval of every said channel, whereby the baseline tracks environmental drift over hours while remaining insensitive to excursions lasting seconds.

Claim 10: The system of Claim 1, wherein the said trailing window of the dispersion registers is fixed in duration and comprises differing numbers of samples for channels of differing update rate.

Claim 11: The system of Claim 1, wherein the said channels update at two or more differing rates, and the composite hazard index is evaluated at a common decision rate using, for each channel updating more slowly than that rate, the most recent reading held between updates.

Claim 12: The system of Claim 1, wherein the said bands of the designated per-channel standardized deviations are bounded at two standard deviations for a warning band and three and one half standard deviations for a critical band.

Claim 13: The system of Claim 1, wherein the fixed limits of the absolute backstop comparator (303) comprise a gas concentration limit expressed as an equivalent against a stated calibration analyte, a temperature limit and a sustained flame condition, and wherein no relative humidity limit and no illuminance limit is comprised in the said fixed limits.

Claim 14: The system of Claim 1, wherein the absolute backstop comparator (303) remains operative during an initialization period preceding the population of the baseline and dispersion registers, whereby fixed-limit protection remains operative during initialization.

Claim 15: The system of Claim 1, wherein the system is configured to detect failure of a sensing channel, being a failure to update within a predetermined number of expected intervals or the return of a value outside the physical range of that channel, and upon such detection to enter a latched sensor-fault state identifying the failed channel, to indicate that state through the indicating means and to record it, and to exclude that channel from the composite hazard index by setting its standardized deviation to zero and removing every pairwise product having that deviation as an operand, the sensor-fault state persisting until cleared by an operator so that the exclusion is not treated as an absence of hazard evidence.

Claim 16: The system of Claim 1, further comprising a recording means (502) configured to store, for each evaluation of the composite hazard index, the individual terms and the pairwise products separately from the index itself, whereby the parameter pair responsible for a subsequent actuation is identifiable after the event.

Claim 17 (Independent): Hazard Evaluation and Arbitration Unit A hazard evaluation and arbitration unit for an environmental protection system, the unit comprising a processor, a memory and an actuator interface, and being configured to:

(a) receive, from at least four sensing channels responsive to measurands in distinct physical domains, a reading per channel, and to maintain for each channel a baseline register holding a running average of that channel and a dispersion register holding a dispersion estimate of that channel over a trailing window;

(b) inhibit updating of the dispersion registers while any standardized deviation derived therefrom exceeds an excursion entry value, and inhibit updating of the baseline registers while any mitigation actuator is commanded above its rest state;

(c) form, for each channel, at least one standardized deviation as the positive part of a signed difference between the reading and the baseline divided by the dispersion estimate of that channel, the sign being taken in the hazard direction of the channel concerned, whereby deviations of channels of differing physical domain and differing hazard direction are rendered dimensionless and mutually comparable, and form a composite hazard index as the sum of weighted individual standardized deviations and weighted pairwise products of selected standardized deviations produced by the normalization stage, at least one coefficient of the said pairwise products being non-zero;

(d) assign the composite hazard index to one of at least three bands, and assign to one of at least three bands, for each said channel, at least one standardized deviation of that channel designated as its per-channel deviation, and compare one or more selected readings against respective fixed limits not modifiable by the said registers; and

(e) command, through the actuator interface, a ventilation actuator of the protected environment to one of at least three levels corresponding to the highest band called for by the composite hazard index, by any said designated per-channel standardized deviation, or by the said comparison, no one of said three paths being capable of suppressing another, the said coefficients and the said bands of the composite hazard index being configured such that the composite hazard index is capable of entering a band above the lowest band while every said designated per-channel standardized deviation remains below its own warning boundary and no reading that is subject to a fixed limit has reached that limit, whereby the physical state of the protected environment is altered.

Claim 18: The unit of Claim 17, wherein the running average held in each said baseline register is an exponentially weighted moving average, the said trailing window is fixed in duration and comprises differing numbers of samples for channels of differing update rate, and the composite hazard index is formed at a common decision rate using the most recent reading held between updates for any channel updating more slowly than that rate.

Claim 19: The unit of Claim 17, wherein the inhibition of Claim 17(b) is structural and is not subject to revision by any calibration of the said weights, coefficients or band boundaries.

Claim 20: The unit of Claim 17, wherein the said actuator interface commands a ventilation duty cycle and an exhaust aperture position, each selected from three or more discrete values according to the band adopted.

Claim 21: The system of Claim 1, wherein in operation the arbiter adopts a band above the lowest band in response to the composite hazard index while every said designated per-channel standardized deviation remains below its own warning boundary and no reading that is subject to a fixed limit has reached that limit.

Claim 22: The system of Claim 1, wherein the running average held in each baseline register (201) is an exponentially weighted moving average.

Claim 23: The system of Claim 1, wherein the weights of the individual standardized deviations are each non-negative and sum to unity and the coefficients of the pairwise products are each non-negative, whereby the composite hazard index is non-negative and monotonically non-decreasing in every standardized deviation.

Claim 24: The system of Claim 1, wherein a first subset of said channels is provided with a said fixed limit and a second subset is not, the channels of the second subset contributing to the composite hazard index and to the per-channel bands only.

Claim 25: The system of Claim 15, wherein failure of the chemical channel or of the optical combustion channel further causes the mitigation actuator to be commanded to an intermediate level until the sensor-fault state is cleared.

Claim 26: The system of Claim 2, wherein for at least one condition of the protected environment, while every designated per-channel standardized deviation remains below its own warning boundary and no reading that is subject to a fixed limit has reached that limit, the sum of the weighted individual standardized deviations alone remains below the first boundary of the composite hazard index, while addition of the sum of the weighted pairwise products causes the composite hazard index to reach or exceed that boundary, whereby the mitigation actuator is commanded above its rest state by the interaction terms and would not be so commanded in the absence of those interaction terms.

11. Technology Readiness Level of the Invention

TRL 1	TRL 2	TRL 3	TRL 4	TRL 5	TRL 6	TRL 7	TRL 8	TRL 9
Principles observed	Concept formed	Proof of concept	Lab validated	Relevant env.	Relevant env. demo	Operational proto.	Qualified	Proven in use
			✓	✓				

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